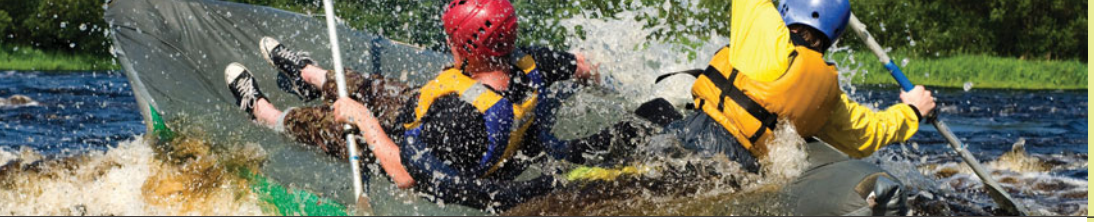
CHAPTER 5

Attitudes

Evaluating and Responding to the Social World





WHAT IS THE BASIS OF PEOPLE'S ATTITUDES TOWARD President Barack Obama? Might how people feel about him affect what they believe about him? What if an attitude is formed based on beliefs that are "disproven"? Let's consider these questions in terms of an issue we hear about frequently in the blogs, as well as legitimate news outlets—is President Obama a Muslim? In analyzing attitudes toward President Obama, the Pew Research Center reports that, as of August 2010, 18 percent of the U.S. population believes that Obama is a Muslim, a new high. How does such a belief get formed? And why does that belief, despite attempts to deny or correct it, apparently have such staying power?

First of all, Obama's well-known personal history has some unusual features. He was born in 1961 in Hawaii to a white American mother, but his biological father was a Muslim from Kenya. Although Obama had little contact with his father during his childhood, the young Barack lived for 4 years with his mother and stepfather in Indonesia, which is the largest Muslim country in the world. For these reasons, people might expect that Obama was introduced early on to the teachings of Islam. On the other hand, when Barack was 10 years old he returned to Hawaii to live with his Christian grandparents, and after that he attended universities on the mainland. As an adult, Obama and his wife went to church and had a close relationship for 20 years with Jeremiah Wright, a Christian preacher in Chicago, although amazingly some say he did this while simultaneously (and secretly) attending a mosque!

The idea that beliefs persist, and continue to be held onto by people—even when strong disconfirmation is provided—is not a new issue to social psychologists. Leon Festinger and colleagues, in their 1956 book, *When Prophecy Fails,* provides us with an inside look at this seeming mystery. In this early investigation of attitudes, Festinger describes a certain Mrs. Keech, a Utah woman of deep faith, who believed that the world was going to end on the morning of December 21, 1954. Festinger details his realization that there was very little that could displace either the woman's or her followers' ardent belief that, indeed, the end of the world was nigh.

This early research revealed several characteristics that are likely to cause people to ignore disconfirming evidence (factual evidence that proves a strongly held belief to be wrong). One such characteristic illustrates our true believer situation rather

CHAPTER OUTLINE

Attitude Formation: How Attitudes Develop

- Classical Conditioning: Learning Based on Association
- Instrumental Conditioning: Rewards for the "Right" Views
- Observational Learning: Learning by Exposure to Others

When and Why Do Attitudes Influence Behavior?

- Role of the Social Context in the Link Between Attitudes and Behavior
- Strength of Attitudes
- Attitude Extremity: Role of Vested Interests
- Attitude Certainty: Importance of Clarity and Correctness
- Role of Personal Experience

EMOTIONS AND ATTITUDE FORMATION
When What the Ad Promises Matches How We Feel

How Do Attitudes Guide Behavior?

- Attitudes Arrived at Through Reasoned Thought
- Attitudes and Spontaneous Behavioral Reactions

The Fine Art of Persuasion: How Attitudes Are Changed

- Persuasion: Communicators, Messages, and Audiences
- The Cognitive Processes Underlying Persuasion

SOCIAL LIFE IN A CONNECTED WORLD
Electronic Word-of-Mouth Marketing and Persuasion

Resisting Persuasion Attempts

- Reactance: Protecting Our Personal Freedom
- Forewarning: Prior Knowledge of Persuasive Intent
- Selective Avoidance of Persuasion Attempts
- Actively Defending Our Attitudes: Counterarguing Against the Competition
- Individual Differences in Resistance to Persuasion
- Ego-Depletion Can Undermine Resistance

Cognitive Dissonance: What Is It and How Do We Manage It?

- Dissonance and Attitude Change: The Effects of Induced Compliance
- Alternative Strategies for Resolving Dissonance
- When Dissonance Is a Tool for Beneficial Changes in Behavior



FIGURE 5.1 How Are Attitudes Toward President Barack Obama Formed?
Do our beliefs (cognitions) shape our attitudes (feelings)? Or, is it the other way around—do our feelings shape our beliefs? Do attitudes change when we are confronted with information that disconfirms our beliefs, or are those beliefs likely to be maintained to the extent that we can find others who share those beliefs?

perfectly: If Mrs. Keech could convince others of her basic premise, then the magnitude of her discomfort following disconfirmation of her belief would be reduced. Indeed, these researchers found that the inevitable disconfirmation of the belief that the world would end was followed by an enthusiastic effort at proselytizing others to join her group. If true believers can find others who provide social support by sharing their beliefs, then the pain of exposure to disconfirming evidence is lessened. As we discuss in this chapter, there is considerable evidence that people hold beliefs that help them make sense of their emotions, even in the face of evidence that strongly disconfirms those beliefs (Boden & Berenbaum, 2010).

Nowadays, with the aid of the Internet, attitude formation can be facilitated from the beginning by the knowledge that other people share one's beliefs. People on the Internet can find each other and begin to build up a store of "evidence" such as Obama's father's religion or his early years in Indonesia, which they collectively agree points to Obama's Muslim identity, even if that evidence is circumstantial at best. And, when additional facts point to Obama's Christian faith, true believers are likely to embrace their belief in his Muslim identity even more strongly! That is, disconfirming evidence can fuel true believers' adherence to their belief, and sharing it with others can further cement that belief in place (see Figure 5.1).



In this chapter we explore the factors that shape the attitudes we hold, and address the key question of whether our attitudes are simply a product of rational thought. We consider how other people affect the attitudes we form, and what happens when we react against their attempts to influence us. How people respond to explicit attempts to persuade them is a complicated issue involving several different processes. We consider when, for example, people closely scrutinize the arguments presented in a message and when communicator credibility is not closely examined (see Figure 5.2 for an amusing take on this issue). We also address the important issue of when and how we manage to persuade ourselves—why our behavior can lead us to change our own attitudes. Along the way we consider whether all attitudes are equal, or if some attitudes are more strongly linked to behavior than others. Lastly, we examine the process by which our attitudes guide our behavior.

Social psychologists use the term **attitude** to refer to people's *evaluation* of almost any aspect of the world (e.g., Olson & Kendrick, 2008; Petty, Wheeler, & Tormala, 2003). People can have favorable or unfavorable reactions to issues, ideas, objects, actions (do you like white water rafting), a specific person (such as Barack Obama) or entire social groups (Muslims). Some attitudes are quite stable and resistant to change, whereas others may be unstable and show considerable variability depending on the situation (Schwarz & Bohner, 2001). We may hold some attitudes with great certainty, while our attitudes toward other objects or issues may be relatively unclear or uncertain (Tormala & Rucker, 2007).

attitude
Evaluation of various aspects of the social world

What is *your* attitude toward the legalization of marijuana, an issue currently on the agenda of many state legislatures—(see Figure 5.3)? Is your attitude toward marijuana

likely to depend on whether you have used it or not? Later in this chapter we consider how our own actions can influence our attitudes (Maio & Thomas, 2007). Does it matter whether you think other people see its use as acceptable or not? What role does consensus—the extent to which we see others as sharing our attitudes—have on the attitudes we hold? Does the fact that this is an issue undergoing social change (see the map of U.S. states that have already or are currently considering legalizing marijuana in Figure 5.3) mean that many people's attitudes are likely to be unstable and subject to change? Does the purpose or how marijuana legalization messages are framed—for the treatment of medical problems or recreational use—matter for the attitudes people hold?

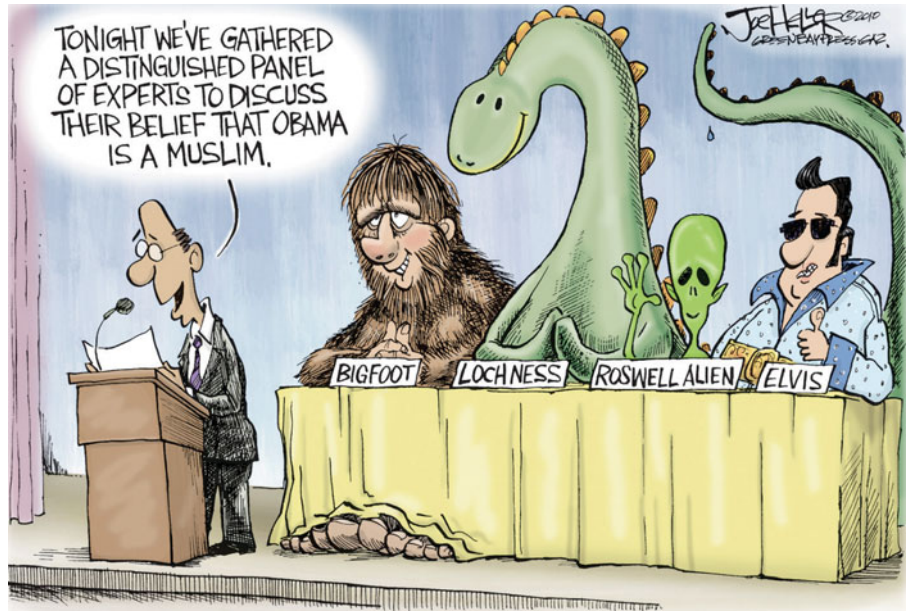


FIGURE 5.2 Why Do So Many People Seem to Agree with This Erroneous Belief?

Public opinion polls in 2010 indicate that 18 percent of the U.S. population agrees with the belief that "President Obama is a Muslim." As this cartoon suggests, perhaps the credibility of the people who support this view should be more closely examined!

The study of attitudes is central to the field of social psychology because attitudes are capable of coloring virtually every aspect of our experience. Even when we do not have strong attitudes toward a specific issue such as the legalization of marijuana, related values can influence what attitudes we form. Let's consider public attitudes toward various scientific issues, specifically the use of human embryonic stem cells. Research findings indicate that attitudes toward such novel issues are shaped by long-term values—religious beliefs predict the formation of these new attitudes—rather than the extent to which the public possesses scientific knowledge on the topic (Ho, Brossard, & Scheufele, 2008). As we saw in Chapter 2, the tendency to evaluate stimuli as positive or negative—something we favor or are against—appears to be an initial step in our efforts to make sense out of the world. In fact, such reactions occur almost immediately, even before we can fully integrate a new stimulus into our previous experience. Responding to a stimulus in terms of our attitudes—on an immediately evaluative basis—produces different brain wave activity than when a response is made on a nonevaluative basis (Crites & Cacioppo, 1996). Our brains operate differently depending on whether we are engaged in rapid evaluative perception or a more thoughtful examination of our world.

In addition, attitudes can influence our thoughts, even if they are not always reflected in our overt behavior. Moreover, while many of our attitudes are **explicit attitudes**—conscious and reportable—other attitudes may be **implicit attitudes**—uncontrollable and perhaps not consciously accessible to us. Consider this explicit versus implicit attitudes distinction as it applies to racial attitudes. Many "color-blind" or self-perceived egalitarian Americans will report positive explicit attitudes toward African Americans. However, they may also display negative involuntary evaluative reactions toward African Americans—implicit attitudes—because it is almost impossible to grow up in the United States without acquiring such negative racial associations (Fazio & Olson, 2003). Furthermore, such implicit attitudes have consequences for important outcomes such as juror decision making when the defendant is African American (Goff, Eberhardt, Williams, & Jackson, 2008).

While social psychologists can learn people's attitudes about many objects from their conscious reports of the thoughts and feelings they have about them, another approach is required if we want to learn someone's implicit attitudes—that is, attitudes they may be either unwilling or unable to report. A method for assessing these is the *Implicit Association Test* (IAT; Greenwald, McGhee, & Schwarz, 1998). The IAT is based on the fact that we

explicit attitudes
Consciously accessible attitudes that are controllable and easy to report.

implicit attitudes
Unconscious associations between objects and evaluative responses.

may associate various social objects more or less readily with positive or negative descriptive words. When there is a close association between a social group—say, Canadians—and some evaluative word such as "polite," one's reaction in identifying this connection is faster than if the social object was paired with a word that one did not readily associate with Canadians, perhaps "rude." Quicker reactions to positive objects and one social group over another can reflect differential valuing of that group. Consider the gender gap in wages that continues to exist today. Might it be that this is due, in part, to the valued attribute of "money" being automatically associated with men versus women? Recent research by Williams, Paluck, and Spencer-Rodgers (2010) using the IAT obtained evidence that male references (e.g., man, son, husband) were automatically associated with wealth-related terms (e.g., rich, cash, paycheck) as indicated by faster response latencies to those pairings than with female references (e.g., mother, aunt, daughter). If you dare, the website http://implicit.harvard.edu/implicit offers a wide-ranging set of IATs about groups that you can take to learn your implicit attitudes about those groups.



Before doing so, though, consider one warning: Although the IAT is viewed by some investigators as an important way to "get inside your head," a criticism that has been leveled at this test is that it really assesses commonly known connections between social groups and various adjectives, even though the respondent might not actually endorse the validity of those connections. That is, one might be fully aware of a common negative stereotype regarding a particular social group, but not personally concur with that negative belief. Consider the possibility raised by Arkes and Tetlock (2004). Because well-known African American leader Jesse Jackson is likely to have knowledge of the negative stereotypic attributes associated with African Americans—he might

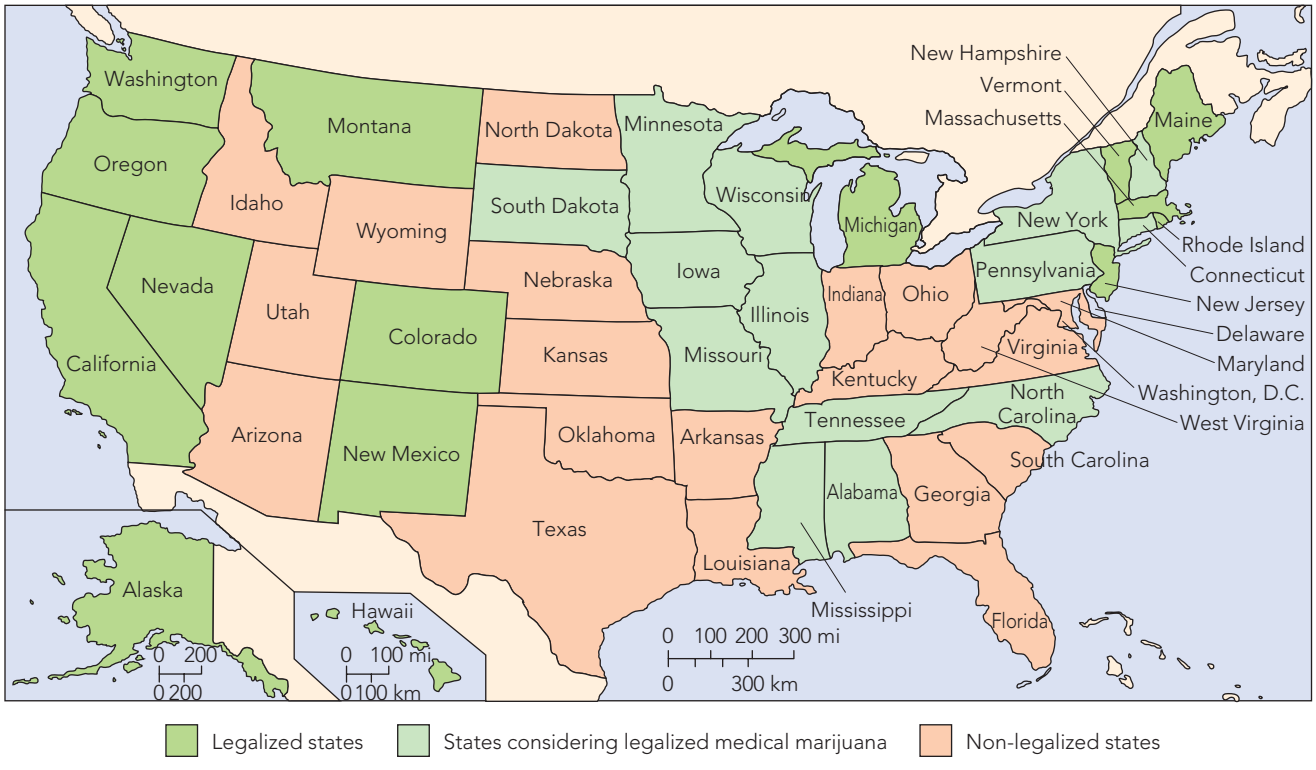


FIGURE 5.3 Marijuana Attitudes: To Support Legalization or Not

As of 2010, 15 U.S. states have legalized the use of marijuana for medical purposes, and another 15 states are considering legislation to do so. What factors influence people's attitudes toward this substance?

"fail" the IAT! That is, this measure might indicate that he holds negative attitudes toward his own group, African Americans. This implies that such implicit measures may be assessing familiarity with the culture rather than an individual's *actual* attitudes. Moreover, research has revealed that the IAT is susceptible to deliberate faking (Fiedler, Messner, & Bluemke, 2006) and that it becomes easier to do so as people gain experience with the IAT (Blair, 2002). Thus, the meaning of IAT scores remains controversial (Gawronski, LeBel, & Peters, 2007). Taken together, though, it is clear from a meta-analytic review of research on implicit and explicit attitudes that they reflect distinct evaluations of the world around us, and implicit attitudes can predict some behaviors better than explicit attitude measures (Greenwald, Poehlman, Uhlmann, & Banaji, 2009).



FIGURE 5.4 Attitudes Toward Celebrities Predict Behaviors Reflecting Interest in Their Lives

When people hold positive attitudes toward particular celebrities (from left to right: Bristol Palin and Paris Hilton), they are likely to enjoy hearing about events in their lives, follow their postings on twitter, and generally attend to information about them.

Another reason that social psychologists view attitudes as important is that they *do* often affect our behavior. This is especially likely to be true when attitudes are strong and accessible (Ajzen, 2001; Bizer, Tormala, Rucker, & Petty, 2006; Fazio, 2000). What is your attitude toward Bristol Palin and Paris Hilton? If positive, you may enjoy hearing about events in their lives on *Entertainment Tonight* as shown in Figure 5.4. Do you like reality TV? If so, we might safely predict that you will probably choose to watch *Survivor*, *Sarah Palin's Alaska*, *Dancing with the Stars*, or *The Apprentice*.

Because attitudes can also affect important behavioral choices that have long-term consequences, it is important to understand how thought processes influence attitude-based decision making. Suppose you receive an e-mail from your student health services office encouraging you to get the flu shot this fall in order to ward off potentially catching the flu in the future? What factors are likely to influence your choice to do so or not? Because people differ in the extent to which they give weight to future consequences when they make such decisions, this might affect how information about getting vaccinated is processed and therefore attitude-based decisions. Morison, Cozzolino, and Orbell (2010) proposed the model shown in Figure 5.5 where considering future consequences should lead to more positive thoughts about a message concerning the vaccine's benefits and risks, and these thoughts should predict attitudes toward the vaccine. To test their model, these investigators first assessed parents' tendencies to consider future consequences of their decisions, and then gave them

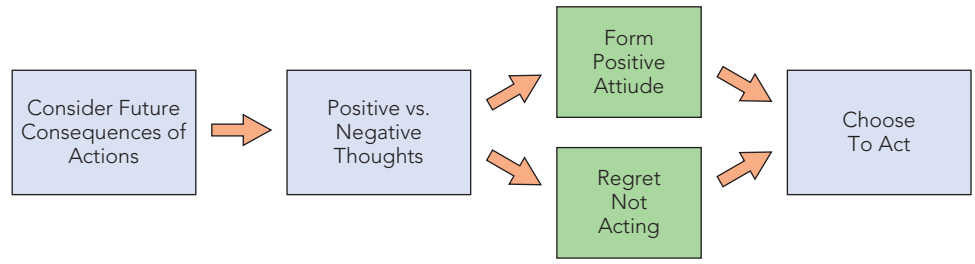


FIGURE 5.5 Factors That Influence Attitudes and Medical Decision-Making

People who consider the future consequences of their actions reported more positive than negative thoughts about a vaccine after reading balanced information about its potential benefits and risks, and this predicted their attitudes about the vaccine and the extent to which regret for not acting was anticipated—which then predicted the decision to have their daughter vaccinated for the human papilloma virus (an important cause of cervical cancer in adult women). (Source: Based on research by Morison, Cozzolino, & Orbell, 2010).

balanced information concerning the benefits and risks of agreeing to have their daughters vaccinated for the human papilloma virus (which causes cervical cancer in women). After reading the information about the virus and vaccine, parents listed their thoughts about it, which were later coded as positive or negative. Then, attitudes toward the vaccine were measured, as was anticipated regret if they did not have their daughter vaccinated and she gets the virus in the future. Finally, the parents' agreement to have their daughter vaccinated was assessed. Results supported the model: Parents who think more about future consequences of their actions generated more positive thoughts (relative to negative thoughts) about the vaccination, which in turn predicted more positive attitudes toward the vaccine and greater anticipated regret of not doing so—and these both fed into choosing to have their daughter vaccinated within the next year. So, sometimes attitudes are formed on the basis of careful consideration of the information and, once those attitudes are formed, they can predict behavior in important domains such as medical decision making.

In this chapter, we consider many influences on *attitude formation.* After doing so, we consider in-depth a question we have already raised: When do attitudes influence behavior and when do they not? Then, we turn to the important question of how attitudes are changed—the process of *persuasion.* We also examine some reasons *why* attitudes are often resistant to change. Finally, we consider the intriguing fact that on some occasions our own actions shape our attitudes rather than vice versa. The process that underlies such effects is known as *cognitive dissonance*, and it has fascinating implications not just for attitude change, but for many aspects of social behavior as well.

Attitude Formation: How Attitudes Develop

How do you feel about each of the following: people who cover their bodies in tattoos, telemarketers, the TV programs *Modern Family*, *Lost*, and *Lie to Me*, sushi, the police, dancing, cats, and people who talk on their cell phones while driving? Most people have attitudes about these issues and objects. But where, precisely, did these views come from? Did you acquire them as a result of your own experiences with each, from other people with whom you have had personal contact, or through exposure via the media? Are your attitudes toward these objects constant across time, or are they flexible and likely to change as conditions do? One important means by which our attitudes develop is through the process of **social learning.** In other words, many of our views are acquired in situations where we interact with others, or simply observe their behavior. Such learning occurs through several processes, which are outlined below.

social learning
The process through which we acquire new information, forms of behavior, or attitudes from other people.

Classical Conditioning: Learning Based on Association

It is a basic principle of psychology that when a stimulus that is capable of evoking a response—the **unconditioned stimulus**—regularly precedes another neutral stimulus, the one that occurs first can become a signal for the second—the **conditioned stimulus.** Advertisers and other persuasion agents have considerable expertise in using this principle to create positive attitudes toward their products. Although tricky in the details, it is actually a fairly straightforward method for creating attitudes. First, you need to know what your potential audience already responds positively toward (what to use as the unconditioned stimulus). If you are marketing a new beer, and your target audience is young adult males, you might safely assume that attractive young women will produce a positive response. Second, you need to pair your product repeatedly (the formerly neutral or conditioned stimulus—say, your beer logo) with images of beautiful women and,

classical conditioning
A basic form of learning in which one stimulus, initially neutral, acquires the capacity to evoke reactions through repeated pairing with another stimulus. In a sense, one stimulus becomes a signal for the presentation or occurrence of the other.

unconditioned stimulus
A stimulus that evokes a positive or negative response without substantial learning.

conditioned stimulus
The stimulus that comes to stand for or signal a prior unconditioned stimulus.

before long, positive attitudes will be formed toward your new beer! As shown in Figure 5.6, many alcohol manufacturers have used this principle to beneficially affect sales of its product.

Such classical conditioning can affect attitudes via two pathways: the direct and indirect route (Sweldens, van Osselaer, & Janiszewski, 2010). The more generally effective and typical method used—*the direct route*—can be seen in the advertisement. That is, positive stimuli (e.g., lots of *different* women) are repeatedly paired with the product, with the aim being to directly transfer the affect felt toward them to the brand. However, by pairing a *specific* celebrity endorser who is already liked by the target audience with the new brand, a memory link between the two can be established. In this case—the indirect route—the idea is that following repeatedly presenting that specific celebrity with the product, then whenever that celebrity is thought of, the product too will come to mind. Think here of Michael Jordan; does Nike come to mind more rapidly for you? For this indirect conditioning process to work, people need not be aware that this memory link is being formed, but they do need to feel positively toward the unconditioned stimulus—that is, that particular celebrity (Stahl, Unkelbach, & Corneille, 2009). Figure 5.7 presents a recent example of this indirect conditioning approach and advertising.



FIGURE 5.6 Classical Conditioning of Attitudes—The Direct Route

Initially people may be neutral toward this brand's label. However after repeatedly pairing this product's logo with an "unconditioned stimulus" of various women who are attractive to the targeted group of young males, seeing the beer logo may come to elicit positive attitudes on its own.

Not only can classical conditioning contribute to shaping our attitudes—it can do so even though we are not aware of the stimuli that serve as the basis for this kind of conditioning. For instance, in one experiment (Krosnick, Betz, Jussim, & Lynn, 1992), students saw photos of a stranger engaged in routine daily activities such as shopping in a grocery store or walking into her apartment. While these photos were shown, other photos known to induce either positive or negative feelings were exposed for very brief periods of time—so brief that participants were not aware of their presence. Participants who were nonconsciously exposed to photos that induced positive feelings (e.g., a newlywed couple, people playing cards and laughing) liked the stranger better than participants who had been exposed to photos that nonconsciously induce negative feelings (e.g., open-heart surgery, a werewolf). Even though participants were not aware that they had been exposed to the second group of photos because they were presented very briefly, the photos did significantly influence the attitudes that were formed toward the stranger. Those exposed to the positive photos reported more favorable attitudes toward this person than those exposed to the negative photos. These findings suggest that attitudes can be influenced by **subliminal conditioning**—classical conditioning that occurs in the absence of conscious awareness of the stimuli involved.



FIGURE 5.7 Classical Conditioning of Attitudes—The Indirect Route

The manufacturers of these watches hope that by repeatedly pairing Tiger Woods with their product, a memory link between that celebrity and the product will be created. If the link formed in memory is sufficiently strong, then whenever consumers think of that celebrity, their watch brand name will come to mind.

Indeed, **mere exposure**—having seen an object before, but too rapidly to remember having seen it—can result in attitude formation (Bornstein & D'Agostino, 1992). We know that this is a case of subliminal conditioning because patients with advanced Alzheimer's disease—who therefore cannot remember seeing the stimuli—show evidence of having formed new attitudes as a result of mere exposure (Winograd, Goldstein, Monarch, Peluso, & Goldman, 1999). It is also the case that even when we can remember being exposed to information, its mere repetition creates

a sense of familiarity and results in more positive attitudes. Moons, Mackie, and Garcia-Marques (2009) refer to this as the **illusion of truth effect.** The studies by these researchers revealed that more positive attitudes developed following exposure to *either* weak or strong arguments—as long as little detailed message processing occurred. Although this has substantial implications for the likely impact of advertising on the attitudes we form—as a result of merely hearing the message repeated—it is good to know that this effect *can* be overcome when people are motivated to and able to process extensively the message.

Once formed, such attitudes can influence behavior—even when those attitudes are inconsistent with how we are explicitly expected to behave. Consider the child whose attitudes toward an ethnic or religious group such as Arabs or Muslims have been classically conditioned to be negative, and who later are placed in a classroom where such negative attitudes are non-normative (i.e., they are deemed unacceptable). Research conducted in Switzerland by Falomir-Pichastor, Munoz-Rojas, Invernizzi, and Mugny (2004) has revealed that, as shown in Figure 5.8, when the norms are anti-discriminatory, if feelings of threat from that "outsider" group are low, then the expression of prejudice can be reduced. When, however, feelings of threat are high, then the child is likely to continue to show prejudice even when the norms are anti-discriminatory. This research illustrates that only when threat is absent are attempts to change negative responses effective using explicit norms.

subliminal conditioning
Classical conditioning of attitudes by exposure to stimuli that are below individuals' threshold of conscious awareness.

mere exposure
By having seen before, but not necessarily remembering having done so, attitudes toward an object can be formed.

illusion of truth effect
The mere repetition of information creates a sense of familiarity and more positive attitudes.

instrumental conditioning
A basic form of learning in which responses that lead to positive outcomes or which permit avoidance of negative outcomes are strengthened.

social networks
Composed of individuals with whom we have interpersonal relationships and interact with on a regular basis.

Instrumental Conditioning: Rewards for the "Right" Views

When we asked you earlier to think about your attitudes toward marijuana, some of you may have thought immediately "Oh, that's wrong!" This is because most children have been repeatedly praised or rewarded by their parents and teachers ("just say no" programs) for stating such views. As a result, individuals learn which views are seen as the "correct" attitudes to hold—because of the rewards received for voicing those attitudes by the people they identify with and want to be accepted by. Attitudes that are followed by positive outcomes tend to be strengthened and are likely to be repeated, whereas attitudes that are followed by negative outcomes are weakened so their likelihood of being expressed again is reduced. Thus, another way in which attitudes are acquired is through the process of **instrumental conditioning**—differential rewards and punishments. Sometimes the conditioning process is rather subtle, with the reward being psychological acceptance—by rewarding children with smiles, approval, or hugs for stating the "right" views. Because of this form of conditioning, until the teen years—when peer influences become especially strong—most children express political, religious, and social views that are highly similar to those of their parents and other family members (Oskamp & Schultz, 2005).

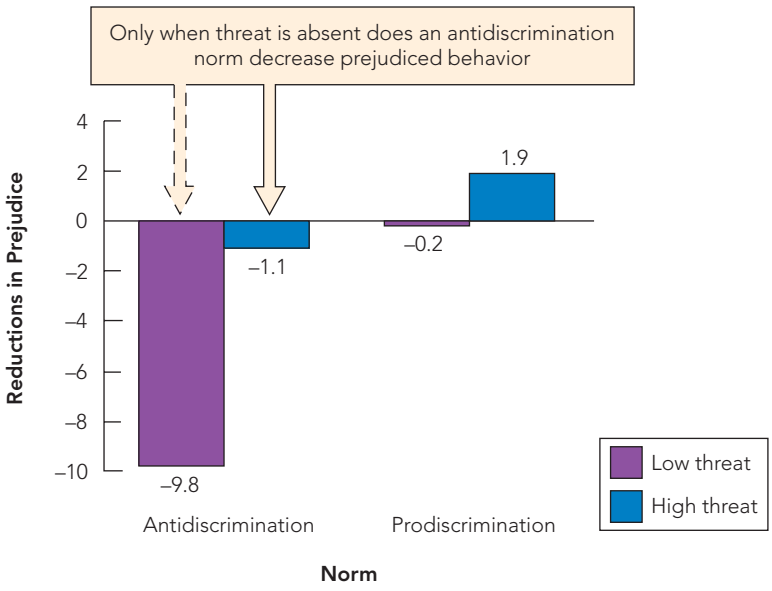


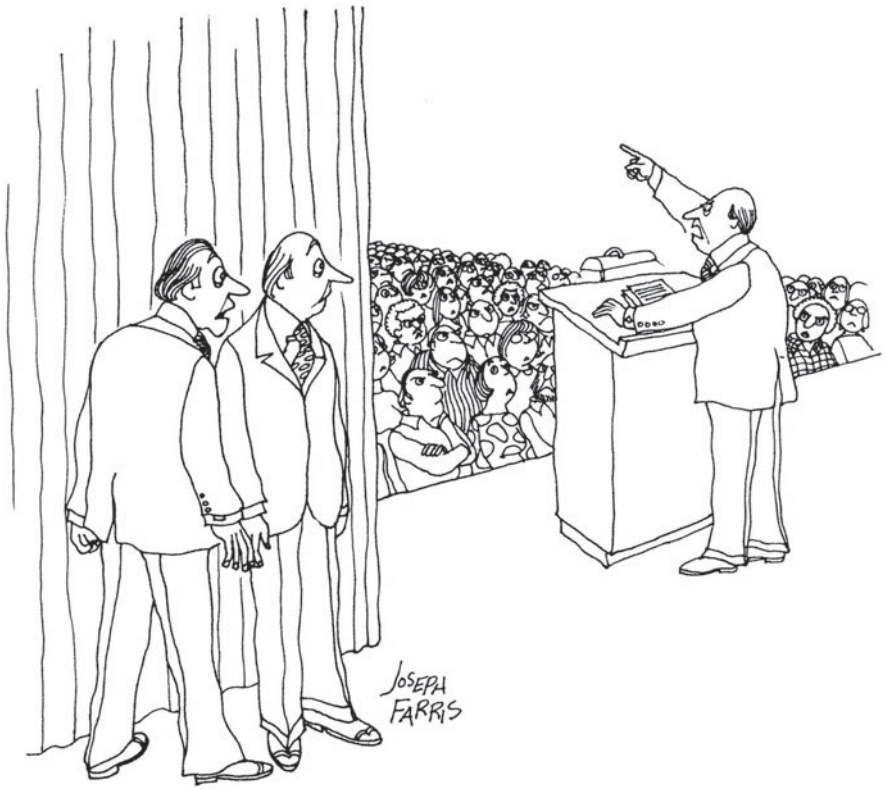
FIGURE 5.8 Feelings of Threat Can Result in Prejudiced Action, Even When Norms Are Anti-Discriminatory
In this study, an anti-discrimination norm against showing prejudice toward foreigners was only effective at reducing favoritism *toward members of their own group when people were feeling* little *threat. But, if a pro-discrimination norm is present, people discriminate by showing favoritism toward their own group members regardless of feelings of threat. (Source: Based on research by Falomir-Pichastor, Munoz-Rojas, Invernizzi, & Mugny, 2004).*

What happens when we find ourselves in a new context where our prior attitudes may or may not be supported? Part of the college experience involves leaving behind our families and high school friends and entering new **social networks**—sets of individuals with whom we interact on a regular basis (Eaton, Majka, & Visser, 2008). The new networks (e.g., new sorority or fraternity) we find ourselves in may contain individuals who share our attitudes toward important social issues, or they may be composed of individuals holding diverse and diverging attitudes toward

those issues. Do new attitudes form as we enter new networks in order to garner rewards from agreeing with others who are newly important to us? To investigate this issue, Levitan and Visser (2009) assessed the political attitudes of students at the University of Chicago when they arrived on campus and determined over the course of the next 2 months the networks the students became part of, and how close the students felt toward each new network member. This allowed the researchers to determine the effect of attitude diversity among these new peers on students' political attitudes. Those students who entered networks with more diverse attitudes toward affirmative action exhibited greater change in their attitudes over the 2-month period. These results suggest that new social networks can be quite influential—particularly when they introduce new strong arguments not previously encountered (Levitan & Visser, 2008). The desire to fit in with others and be rewarded for holding the same attitudes can be a powerful motivator of attitude formation and change.

It is also the case that people may be consciously aware that different groups they are members of will reward (or punish) them for expressing support for particular attitude positions. Rather than being influenced to change our attitudes, we may find ourselves expressing one view on a topic to one audience and another view to a different audience. Indeed, as the cartoon in Figure 5.9 suggests, elections are sometimes won or lost on a candidate's success at delivering the right view to the right audience! Fortunately, for most of us, not only is our every word not recorded, with the possibility of those words being replayed to another audience with a different view, but our potentially incompatible audiences tend to remain physically separated. What this means is that we are less likely than politicians to be caught expressing different attitudes to different audiences!

One way that social psychologists assess the extent to which people's reported attitudes depend on the expected audience is by varying who might learn of their attitude position. For example, people seeking membership in a fraternity or sorority (e.g., pledges) express different attitudes about other fraternities and sororities depending on whether they believe their attitudes will remain private or they think that the powerful members of their group who will be controlling their admittance will learn of the attitude position they advocated (Noel, Wann, & Branscombe, 1995). When those who are attempting to gain membership in an organization believe that other members will learn of "their attitudes," they derogate other fraternities or sororities as a means of communicating to decision makers that the particular organization they want to be admitted to is seen as the most desirable. Yet, when they believe their attitude responses will be private, they do not derogate other fraternities or sororities. Thus, both the attitudes we form and our attitude expression can depend on the rewards we have received in the past and those we expect to receive in the future for expressing particular attitudes.



"Good God! He's giving the white-collar voters' speech to the blue collars."

FIGURE 5.9 Expressing Different Attitudes to Different Audiences
To gain rewards, politicians often tailor their message to match those of their audience. Disaster can strike when the wrong audience gets the wrong message!

Observational Learning: Learning by Exposure to Others

A third means by which attitudes are formed can operate even when direct rewards for acquiring or expressing those attitudes are absent. This process is **observational learning,** and it occurs when individuals acquire attitudes or behaviors simply by observing others (Bandura, 1997). For example, people acquire attitudes toward many topics and objects by exposure to advertising—where we see "people like us" or "people like we want to become" acting positively or negatively toward different kinds of objects or issues. Just think how much observational learning most of us are doing as we watch television!

Why do people often adopt the attitudes that they hear others express, or acquire the behaviors they observe in others? One answer involves the mechanism of **social comparison**—our tendency to compare ourselves with others in order to determine whether our view of social reality is correct or not (Festinger, 1954). That is, to the extent that our views agree with those of others, we tend to conclude that our ideas and attitudes are accurate; after all, if others hold the same views, these views *must* be right! But are we equally likely to adopt all others' attitudes, or does it depend on our relationship to those others?

People often adjust their attitudes so as to hold views closer to those of others who they value and identify with—their **reference groups.** For example, Terry and Hogg (1996) found that the adoption of favorable attitudes toward wearing sunscreen depended on the extent to which the respondents identified with the group advocating this change. As a result of observing the attitudes held by others who we identify with, new attitudes can be formed.

observational learning
A basic form of learning in which individuals acquire new forms of behavior as a result of observing others.

social comparison
The process through which we compare ourselves to others to determine whether our view of social reality is, or is not, correct.

reference groups
Groups of people with whom we identify and whose opinions we value.

Consider how this could affect the attitudes you form toward a new social group with whom you have personally had no contact. Imagine that you heard someone you like and respect expressing negative views toward this group. Would this influence your attitudes? While it might be tempting to say "Absolutely not!", research findings indicate that hearing others whom we see as similar to ourselves state negative views about a group can lead us to adopt similar attitudes—without ever meeting any members of that group (e.g., Maio, Esses, & Bell, 1994; Terry, Hogg, & Duck, 1999). In such cases, attitudes are being shaped by our own desire to be similar to people we like. Now imagine that you heard someone you dislike and see as dissimilar to yourself expressing negative views toward this group. In this case, you might be less influenced by this person's attitude position. People are not troubled by disagreement with, and in fact expect to hold different attitudes from, people whom they categorize as different from themselves; it is, however, uncomfortable to differ on important attitudes from people who we see as similar to ourselves and therefore with whom we expect to agree (Turner, 1991).

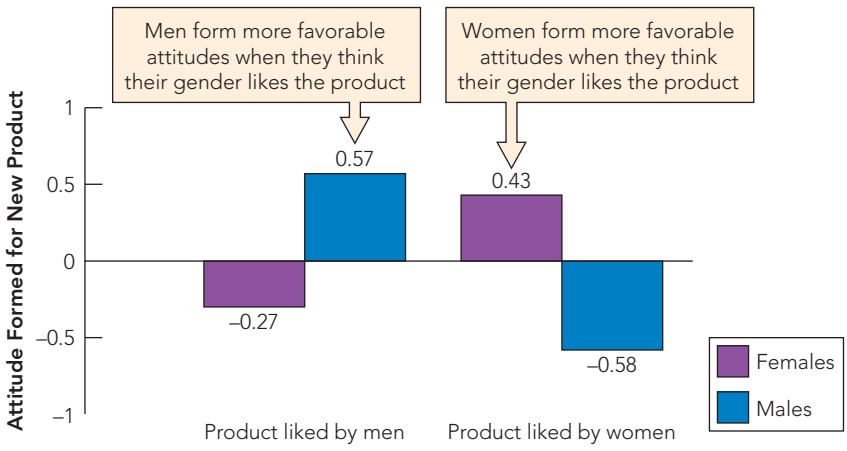


FIGURE 5.10 Attitude Formation Among Those Who Are Highly Identified with Their Gender Group

Men formed more positive attitudes toward the new product when they thought other men liked it, but women formed more positive attitudes toward the product when they thought other women liked it. (Source: Based on data in Fleming & Petty, 2000).

Not only are people differentially influenced by others' attitude positions depending on how much they identify with those others, they also *expect* to be influenced by other people's attitude positions differentially depending on how much they identify with those others. When a message concerning safe sex and AIDS prevention was created for university students, those who identified with their university's student group believed that they would be personally influenced by the position advocated in the message, whereas those who were low in identification with their university's student group did not expect to be personally influenced by the message (Duck, Hogg, & Terry, 1999). Thus, when we identify with a

group, we expect to be influenced by those others and, in fact, are likely to take on the attitudes that are perceived to be normative for that group.

To see this process in action, suppose you were exposed to a new product you have never encountered before. How might the identity relevance of the message influence the attitude you form? To address this question, Fleming and Petty (2000) first selected students to participate in the study who were either high or low in identification with their gender group. Then, they introduced a new snack product ("Snickerdoodles") to men and women as either "women's favorite snack food" or "men's favorite snack food." As Figure 5.10 illustrates, among those who were highly identified with their gender group, a more favorable attitude toward this product was formed when the message was framed in terms of their own group liking that food. In contrast, among those low in identification with their gender group, no differences in the attitudes they formed toward the new food was found as a function of which group was said to favor that food. These findings indicate that the attitudes we form are indeed strongly influenced by our identification with various groups and our perception of what attitudes are held by members of those groups.



KEY POINTS

- **Attitudes** can reflect evaluations of any aspect of the world. Attitudes help us understand people's responses to new stimuli. Attitudes toward new topics can be shaped by long-term values, including religious beliefs.
- Attitudes can be **explicit**—conscious and easy to report—or **implicit**—which implies they are uncontrollable and potentially not consciously accessible. The *Implicit Association Test* is often used to assess whether the associations people have between a group or object are positive or negative.
- Attitudes are acquired from other people through **social learning** processes. Such learning can involve **classical conditioning, instrumental conditioning,** or **observational learning.**
- Attitudes can be classically conditioned even without our awareness—via **subliminal conditioning** and **mere exposure.**
- Attitudes that are acquired through instrumental conditioning stem from differential rewards and punishments for adopting particular views. Attitudes shift as people enter new **social networks** composed of individuals who hold diverging attitudes.
- Because we compare ourselves with others to determine whether our view of social reality is correct or not, we often adopt the attitudes that others hold. As a result of the process of **social comparison,** we tend to adopt the attitude position of those we see as similar to ourselves but not of those we see as dissimilar.
- When we identify with a group, we expect to be influenced by messages that are aimed at our group. We do not expect to be influenced when we do not identify with the group to which the attitude-relevant message is aimed.

When and Why Do Attitudes Influence Behavior?

So far we have considered the processes responsible for the attitudes we form. But we haven't addressed another important question: Do attitudes predict behavior? This question was first addressed more than 70 years ago in a classic study by LaPiere (1934). To determine whether people with negative attitudes toward a specific social group would in fact act in line with their attitudes, he spent 2 years traveling around the United States with a young Chinese couple. Along the way, they stopped at 184 restaurants and 66 hotels and motels. In the majority of the cases, they were treated courteously; in fact, they were refused service only once. After their travels were completed, LaPiere wrote to all the businesses where he and the Chinese couple had stayed or dined, asking whether they would

or would not offer service to Chinese visitors. The results were startling: 92 percent of the restaurants and 91 percent of the hotels that responded said no to Chinese customers!

These results seemed to indicate that there is often a sizeable gap between attitudes and behavior—that is, what a person says and what that person actually does when confronted with the object of that attitude may be quite different. Does this mean that attitudes don't predict behavior? Not necessarily. To understand why attitudes might not straightforwardly predict behavior, we need to recognize that there are various norms that can affect the likelihood of discriminatory behavior. So even the most prejudiced people will not always act on their attitudes—when there are strong situational pressures to do otherwise. Likewise, there are social conditions under which people who do not think of themselves as prejudiced may find themselves discriminating against others based on their group membership. Let's consider now how the social context can affect the link between attitudes and behavior.

Role of the Social Context in the Link Between Attitudes and Behavior

You have probably experienced a gap between your own attitudes and behavior on many occasions—this is because the social context can directly affect the attitude–behavior connection. For instance, what would you say if one of your friends shows you a new tattoo of which he or she is proud and asks for your opinion? Would you state that you do *not* like it, if that was your view? The chances are quite good that you would try to avoid hurting your friend's feelings so you might even say you *like* it even though your attitude is negative. In such cases, we are clearly aware of our conscious choice not to act on our "true" attitude. As this example illustrates, depending on the degree to which the action has social consequences or not, attitudes may be differentially related to behavior. In contrast to your attitude–behavior inconsistency in responding to your friend's tattoo, your attitude might be a very good predictor of whether *you* would get a tattoo or not.

Because of the important role that the social context plays in determining when attitudes and behavior will be related, recent research has focused on the factors that determine *when* consistency can be expected, as well as the issue of *how* attitudes influence behavior. Several factors determine the extent to which attitudes and behavior correspond, with aspects of the situation influencing the extent to which attitudes determine behavior. In addition, features of the attitudes themselves are also important—for example, how *certain* you are of your own attitude. Attitudes that we hold with greater certainty are more strongly linked to behavior (Tormala & Petty, 2004) compared to attitudes about which we feel some uncertainty. Indeed, when people are induced to think that their attitudes are stable across time, they feel more certain about those attitudes and are more likely to act on them (Petrocelli, Clarkson, Tormala, & Hendrix, 2010). It is well known that older people are often more certain of their attitudes than are young people. Recent research suggests that this is partly due to older people placing greater value on "standing firm" or being resolute in the attitude positions they adopt, and for this reason they tend to show greater attitude–behavior consistency compared to younger people (Eaton, Visser, Krosnick, & Anand, 2009).

Have you ever been worried about what others would think of you if you expressed your "true" attitude toward an issue? If so, you will understand the dilemma that Stanford University students experienced in a study conducted by Miller and Morrison (2009). The private attitudes of those students toward heavy alcohol consumption were relatively negative. But, they believed that other students' attitudes toward heavy alcohol consumption were more positive than their own (an instance of **pluralistic ignorance,** where we erroneously believe others have attitudes different than ourselves). When these students were randomly assigned to receive information about other Stanford students' alcohol attitudes—that they held either more positive or more negative attitudes than their own—the students differed in how comfortable they felt expressing their attitude about alcohol use with another Stanford student and their likelihood of choosing alcohol policies as a topic for discussion. The students expressed greater comfort discussing campus drinking and chose that topic for

pluralistic ignorance
When we collectively misunderstand what attitudes others hold and believe erroneously that others have different attitudes than us.

discussion more often when they thought other students' attitudes were more pro-alcohol than their own, but they were less willing to do so when they learned other students' attitudes were more negative than their own. This pattern of wanting to express attitudes in the direction of the perceived campus norm but not when our attitudes go against the norm was especially strong for students who identified highly with their student group.

Strength of Attitudes

Consider the following situation: a large company markets a dangerous product to the public for decades, while internally sharing memos about the addictiveness of the product and how to manipulate that addictiveness. Along the way, an executive of the company has serious moral qualms about the rightness of these actions. Eventually, the concerned employee tips off the news media about these practices and an investigation is begun. The "whistle-blower" is eventually found out and is even sued by his former employer (although the lawsuit that was initiated against him is ultimately dropped).

You may recognize the person and company being described here because these events were ultimately made into a movie, *The Insider*. It was Jeffrey Wigand who blew the whistle on the practices of the tobacco industry in general and his former employer in particular—Brown & Williamson. Why might people take such drastic and potentially risky action (i.e., informing on their employer)? The answer is clear: Such people are passionately committed to the notion that corporations must be honest, especially when there is the potential for damage to the public. Attitudes like these—that are based on moral convictions—can give rise to intense emotion and strongly predict behavior (Mullen & Skitka, 2006). In other words, whether attitudes will predict sustained and potentially costly behavior depends on the strength of the attitudes. Let's consider why attitude strength has this effect.

The term *strength* captures the *extremity* of an attitude (how strong the emotional reaction is), the degree of *certainty* with which an attitude is held (the sense that you know what your attitude is and the feeling that it is the correct position to hold), as well as the extent to which the attitude is based on *personal experience* with the attitude object. These three factors can affect attitude *accessibility* (how easily the attitude comes to mind in various situations), which ultimately determines the extent to which attitudes drive our behavior (Fazio, Ledbetter, & Towles-Schwen, 2000). As shown in Figure 5.11, all of these components of attitude strength are interrelated, and each plays a role in the likelihood that attitudes will be accessible and affect behavior (Petty & Krosnick, 1995). We now take a closer look at each of these important factors.

Attitude Extremity: Role of Vested Interests

Let's consider first attitude *extremity*—the extent to which an individual feels strongly—in one direction or the other—about an issue (Visser, Bizer, & Krosnick, 2006). One of the key determinants of this is what social psychologists term vested interest—the extent to which the attitude is relevant to the concerns of the individual who holds it. This typically amounts to whether the object or issue might have important consequences for this person. The results of many studies indicate that the greater such vested interest, the stronger the impact of the attitude on behavior (Crano, 1995; Visser, Krosnick, & Simmons, 2003). For example, when students

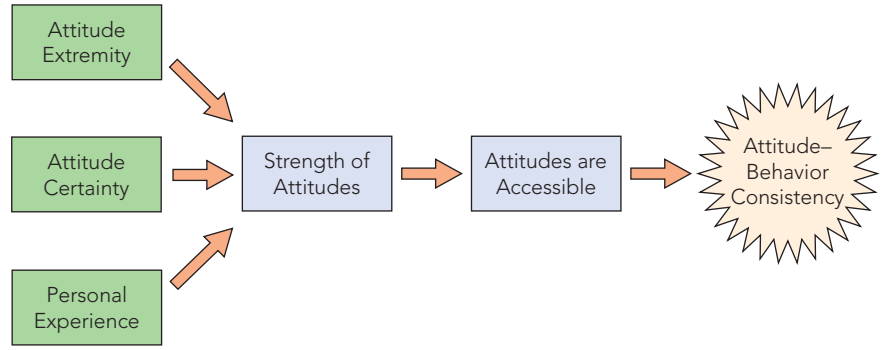


FIGURE 5.11 How Attitude Strength Influences Attitude–Behavior Consistency

Attitudes that are extreme, certain, and formed on the basis of personal experience with the attitude object tend to be strong attitudes, which are more likely to be accessible when a behavioral response is made. Greater attitude–behavior consistency is found when attitudes are strong rather than weak. (Sources: Based on research by Clarkson, Tormala, DeSensi, & Wheeler, 2009; Petrocelli, Tormala, & Rucker, 2007).

at a large university were telephoned and asked if they would participate in a campaign *against* increasing the legal age for drinking alcohol from 18 to 21, their responses depended on whether they would be affected by the policy change or not (Sivacek & Crano, 1982). Students who would be affected by this new law—those younger than 21—have a stronger stake in this issue than those who would not be affected by the law because they were already 21 or would reach this age before the law took effect. Thus, it was predicted that those in the first group—whose interests were at stake—would be much more likely to join a rally against the proposed policy change than those in the second group. This is exactly what happened: While more than 47 percent of those with high vested interest agreed to take part in the campaign, only 12 percent of those in the low vested interest group did so.

Not only do people with a vested interest behave in a way that supports their cause, they are likely to elaborate on arguments that favor their position. By doing so, attitude-consistent thoughts come to mind when an issue is made salient. For example, Haugtvedt and Wegener (1994) found that when participants were asked to consider a nuclear power plant being built in their own state (high personal relevance) they developed more counterarguments against the plan than when the power plant might be potentially built in a distant state (low personal relevance). Thus, attitudes based on vested interest are more likely to be thought about carefully, be resistant to change, and be an accessible guide for behavior.

Recent research findings indicate that vested interests are particularly likely to affect judgments and behavior in the immediate context, whereas abstract values do so when the judgment or behavior is in the distant future (Hunt, Kim, Borgida, & Chaiken, 2010). The issue these researchers tackled was one that has long puzzled those interested in voting, and that Frank (2004) addressed in his book, *What's the Matter with Kansas?* That is, when do people vote their economic self-interests and when do they "apparently act against their economic self-interests" and instead vote in favor of value-based proposals? To test when vested interests are paramount and when they may play a lesser role in behavior, students' material interests were pitted against their egalitarian values. White American students were given a proposal that would be enacted at their university either immediately or in the distant future. It would involve raising tuition by 10 percent in order to restore funds used for recruiting minority students that had been cut. Participants in the immediate condition who would experience the increase *opposed* the proposal, particularly when their own financial strain was high. In effect, they acted on their economic self-interests. In contrast, participants in the distant condition *favored* the proposal to the extent that they had egalitarian social attitudes. This research suggests that vested material interests do affect attitudes and voting when the policy is framed as having an immediate impact, but that for policies framed as having an impact only in the future, people favored and voted based on their values.

Attitude Certainty: Importance of Clarity and Correctness

Research has identified two important components of attitude certainty: attitude clarity—being clear about what one's attitude is—and attitude correctness—feeling one's attitude is the valid or the proper one to hold. Research by Petrocelli, Tormala, and Rucker (2007) provides evidence for the distinction between these two components of attitude certainty by showing how different factors affect them.

To accomplish this task, Petrocelli and colleagues (2007) first determined that their participants felt negatively about a specific attitude issue: requiring students to carry identification cards with them at all times. Then, in order to manipulate the perception of *consensus* concerning their attitude position, half of the participants were given feedback that most other students (89 percent) agreed with their attitude toward the identification card issue, while the other half were told that most other students disagreed (only 11 percent) with them. Although attitude clarity was equivalent in both the high and low consensus conditions, perceived correctness was greater when consensus was high (the 89 percent condition) rather than low (11 percent). When a person learns that others share one's attitudes, it acts as justification for that attitude and thereby increases certainty.

Clarity, the other component of attitude certainty, reflects a lack of ambivalence about an attitude issue. The more often you are asked to report on your attitude, the more it will facilitate clarity and thereby certainty. Repeatedly stating your attitude appears to "work" by increasing your subjective sense that you really *do* know how you feel about an object or issue. When Petrocelli et al. (2007) had their participants express their attitudes toward gun control either several times or only once, attitude certainty differed. Those in the "more expressions" condition had greater certainty about their attitudes toward gun control than those in the "single expression" condition.

What happens when both the clarity and correctness components are varied simultaneously? Returning to the identity card example, Petrocelli et al. (2007) gave students with negative attitudes toward the policy manipulations that were designed to affect both correctness (consensus) and clarity (repeated expression). The students were then given a persuasive message with strong arguments in favor of the policy but against their initial attitudes—why the policy would enhance student safety. More attitude change resulted in the low-clarity case than the high-clarity condition (single vs. repeated expression), and more attitude change occurred in the low-correctness versus the high-correctness condition (low vs. high consensus). Both components of attitude certainty, when they are high, can increase resistance to a persuasive message—each independently contributed to resistance to persuasion.

The social context too is important in assessing the relative effects of attitude clarity and correctness. High clarity will be more predictive of behavior in private but not public contexts—where correctness concerns are likely to be greater. Moreover, when people's attitudes are attacked, successfully resisting those attacks may well increase perceptions of attitude certainty because mounting and expressing counterarguments will increase perceptions of attitude correctness. In terms of attitude–behavior consistency, an attitude that is high on *both* clarity and correctness is most likely to reliably predict behavior in public and in private.

Role of Personal Experience

Depending on how attitudes are formed initially, the link between attitudes and behavior can differ. Considerable evidence indicates that attitudes formed on the basis of direct experience with the object about which we hold a particular attitude can exert stronger effects on behavior than ones formed indirectly. This is because attitudes formed on the basis of direct experience are likely to be stronger and be more likely to come to mind when in the presence of the attitude object (Tormala, Petty, & Brinol, 2002). Similarly, attitudes based on personal relevance are more likely to be elaborated on in terms of supporting arguments, and this makes them resistant to change (Wegener, Petty, Smoak, & Fabrigar, 2004). Consider the difference between having a friend tell you that a particular car model, "Brand X," is a lemon versus having experienced some failures with this brand yourself. When looking at new models of "Brand X," would your friend's opinion even come to mind? Maybe not. Would your own experiences come to mind? Probably. Thus, when you have direct experience with an attitude object it is likely to be quite personally relevant and strong, and your attitude toward it is likely to predict your behavior toward it in the future.

Personal experience is one way to create involvement with an issue, and people who are more involved with an issue and whose values are linked with that issue are more likely to act on their attitudes (Blankenship & Wegener, 2008). For example, when students were asked to consider a novel issue—whether a fictitious country, Tashkentistan, should be allowed to join the European Union—in light of a value of importance to them (e.g., freedom) or in light of a value of little importance (e.g., unity), they spent more time thinking about and elaborating on the message when it involved important values compared to when it did not. This elaboration resulted in stronger attitudes, which in turn guides behavior even in contexts where those attitudes are under attack.

In sum, existing evidence suggests that attitudes really *do* affect behavior (Eagly & Chaiken, 1993; Petty & Krosnick, 1995). However, the strength of this link is strongly determined by a number of different factors. First of all, situational constraints may not permit us to overtly express our attitudes. Second, attitude extremity, which is a function

of whether we have a vested interest in the issue or not, influences whether our attitudes translate into behavior, and this is particularly likely when a message is framed as having an immediate impact rather than one far in the future. Third, attitudes that are clear and experienced as correct are more likely to affect behavior than are those that lack clarity or that we are uncertain about their correctness. Fourth, whether we have personal experience with the attitude object or perceive it as relevant to our important values can affect the accessibility of the attitude, and attitudes that are more accessible are more likely to determine behavior compared to those that are not accessible. For more information on how emotions can influence the attitudes we form about a product, see our special feature, "EMOTIONS AND ATTITUDE FORMATION: When What the Ad Promises Matches How We Feel."

EMOTIONS *and* ATTITUDE FORMATION

When What the Ad Promises Matches How We Feel

How do different emotions affect the attitudes we form toward products that make particular claims about the emotions they will bring? Consider the two advertising photos shown in Figure 5.12. Some vacation ads promise that we will experience much excitement—sailing, playing sports, diving, meeting new people, and so on. We might call these high-arousal positive promises. Other ads for similar locations (i.e., sandy beaches, a warm sea) promise relaxation and peace and quiet—essentially, they offer an opportunity to get "away from it all." We can refer to those as low-arousal positive promise ads. You, no doubt, have had to consider this question when deciding what kind of "spring break" to have—one filled with work, helping people in need in a far-off place, one filled with fun with your student friends in Florida, or one relaxing, catching up on sleep, and reading a good book. Which will it be this year? Perhaps the choice depends on how you are feeling at the time you are forming an opinion about the options and making the decision.

The question that Kim, Park, and Schwarz (2010) asked in their research on this issue was, What are the consequences of experiencing incidental positive feelings that differ in level of arousal at the time we are forming our attitudes toward these vacation products? Of course, we know from much research that people who are in a good mood evaluate all sorts of consumer products more positively than people in bad moods (Schwarz & Clore, 2007). But, positive emotions come in different levels of arousal: high





FIGURE 5.12 Role of Current Emotions in Attitude Formation: When the Ad Promises Excitement and You Want Peace or Vice Versa

When an ad promised either an adventurous or a serene vacation in Japan, participants rated the adventurous product more positively when they were first induced to feel excited rather than peaceful and, conversely, they rated the serene vacation product more positively when they were induced to feel peaceful rather than excited. This was the case when participants' attention was not drawn to their current feelings, which permitted their current feelings to serve as information when forming attitudes toward the vacation products.

(excitement) and low (peaceful). If people use their current emotions as information when forming an attitude toward a new stimulus, then as long as attention is not drawn to their current emotional state, which would discredit its validity for judging the new stimulus, responses to these different vacation products should be affected. However, when attention is drawn to the incidental nature of their current emotional state, it should eliminate use of that emotion as information, and undermine its effect on attitude formation (Schwarz & Clore, 1983).

To test these ideas, participants were first induced to feel either excited or peaceful by describing in detail a life event they had experienced reflecting one of those emotions. After that, as seemingly part of another study entirely, they were given one of two ads for a trip to Japan to evaluate. In one version, the trip was described as "Full of Adventure," with exciting and stimulating activities. In the other version, the trip to Japan was described as "Full of Serenity," featuring peaceful and tranquil activities. After the ad for one of the trip versions was examined, half of the participants were alerted to the potential effect of their own mood on their judgments, or no such awareness cue was provided. Participants were then asked to indicate how much they would like to visit Japan—the advertised destination, and whether taking a trip to this location would be a good decision.

First of all, participants reported feeling equally positive across conditions, but they reported feeling more excited in the excitement writing task condition and more serenity in the serene writing task condition. As predicted, in the absence of an awareness cue reminding participants of the true source of their current feelings and its potential effect on their judgments, the adventurous vacation was rated more favorably when participants felt excited rather than peaceful and the serene vacation was rated more favorably when participants felt peaceful rather than excited. However, when the awareness cue was present, the emotion participants were feeling no longer had an effect on the attitude formed about the vacation options. In a subsequent study, these researchers found the same pattern of effects on participants' expectations that the vacation product they viewed would in fact deliver on its emotional claims were they to actually go on that trip. When current feelings were not discredited as information, then participants believed that the trip to Japan would in fact deliver adventure or serenity, depending on how the participants were feeling (excited or peaceful). So, when you are trying to decide whether "an action will be good for you to do or not," those products that promise to make you feel the specific positive emotion you are currently experiencing may often have an advantage.



KEY POINTS

- Attitudes toward a group, issue, or object do not always directly predict behavior. Rather, there are situational constraints and norms that affect our willingness to express our true attitudes. Concerns about what others, especially those with whom we identify, may think of us can limit the extent to which our attitudes and behavior are consistent.
- People often show **pluralistic ignorance**—erroneously believing that others have different attitudes than themselves. This can limit the extent to which we express our attitudes in public.
- Strong attitudes are ones we are committed to, and we typically have moral values to support them. For this reason, they are more likely to be accessible at the time we take action and are particularly likely to influence behavior.
- Attitude strength subsumes several factors: *extremity, certainty*, and *degree of personal experience*. Those attitudes that are more extreme, certain (both in terms of clarity and perceived correctness), and based on personal experience or important values are more likely to be accessible and guide behavior than are less extreme, unclear, and indirectly formed attitudes.
- Attitude formation can be affected by the specific emotion we are currently feeling when exposed to the object.

How Do Attitudes Guide Behavior?

When it comes to the question of how attitudes guide behavior, it should come as no surprise that researchers have found that there is more than one basic mechanism through which attitudes can shape behavior. We first consider behaviors that are driven by attitudes based on reasoned thought, and then examine the role of attitudes in more spontaneous behavioral responses.

Attitudes Arrived at Through Reasoned Thought

In some situations we give careful, deliberate thought to our attitudes and their implications for our behavior. Insight into the nature of this process is provided by the **theory of reasoned action,** which was later refined and termed the **theory of planned behavior** (Ajzen & Fishbein, 1980). This theoretical view starts with the notion that the decision to engage in a particular behavior is the result of a rational process. Various behavioral options are considered, the consequences or outcomes of each are evaluated, and a decision is reached to act or not to act. That decision is then reflected in *behavioral intentions*, which are often good predictors of whether we will act on our attitudes in a given situation (Ajzen, 1987). Indeed, for a number of behavioral domains—from condom use to engaging in regular exercise—intentions *are* moderately correlated with behavior (Albarracin, Johnson, Fishbein, & Muellerleile, 2001).

Recent research has made it clear that the intention–behavior relationship is even stronger when people have formed a plan for how and when they will translate their intentions into behavior (Frye & Lord, 2009; Webb & Sheeran, 2007). Suppose, for example, that you form the intention to go to the gym to work out. If you develop a plan for *how* you will translate your intention into actual behavior—beginning with setting your alarm, preparing your exercise clothes, and so forth—you will be more likely to succeed at doing so. In my own case, because I formed the intention to walk three mornings a week, I made a commitment to do so with my next-door neighbor. The reason why this is a particularly effective **implementation plan** is that I no longer have to assess whether I *really* want to go out today—in the cold, rain, or whatever, or rely on having my attitude toward getting more exercise be accessible at that time of the morning. As Gollwitzer (1999) has noted, such a plan to implement our intentions is very effective because it involves delegating control of one's behavior to the situation—in my case, my alarm clock beeping and, if that hasn't worked, my neighbor ringing my doorbell!

But, how do you form an intention to change some aspect of your behavior? According to the theory, intentions are determined by two factors: *Attitudes toward the behavior*—people's positive or negative evaluations of performing the behavior (whether they think it will yield positive or negative consequences), and *subjective norms*—people's perceptions of whether others will approve or disapprove of this behavior. A third factor, *perceived behavioral control*—people's appraisals of their ability to perform the behavior—was subsequently added to the theory (Ajzen, 1991). Perhaps a specific example will help illustrate the nature of these ideas.

Suppose an adolescent male is considering joining Facebook. Will he actually take action, find the website, and go through the process of joining up? First, the answer will depend on his intentions, which will be strongly influenced by his attitude toward Facebook. His decision of whether to join or not will also be based on perceived norms and the extent to which he feels able to execute the decision. If the teen believes that becoming a member will be relatively painless and it will make him look more sociable (he has positive attitudes toward the behavior), he also believes that people whose opinions he values will approve of this action (subjective norms), and that he can readily do it (he knows how to access Facebook, upload some photos, and he believes he can control how much of his private data is exposed), his intentions to carry out this action may be quite strong. On the other hand, if he believes that joining Facebook might be dangerous because of the exposure of private data, joining might not really lead to more interaction with friends, or his friends will disapprove of his joining, then his intention to join will be relatively weak. His intentions are more likely to translate into behavior if he formulates a plan for when and how to join (e.g., "On Friday when I get done with school, I'll access the Facebook website and join up"). Of course, even the best of intentions can be thwarted by situational factors (e.g., an emergency that he has to attend to comes up on Friday), but, in general, intentions are an important predictor of behavior.

Reasoned action and planned behavior ideas have been used to predict behavior in many settings, with considerable success. Indeed, research suggests that these theories are useful for predicting such divergent behaviors as soldiers' conduct on the battlefront (Ajzen & Fishbein, 2005) and whether individuals drive a vehicle after they have

theory of reasoned action
A theory suggesting that the decision to engage in a particular behavior is the result of a rational process in which behavioral options are considered, consequences or outcomes of each are evaluated, and a decision is reached to act or not to act. That decision is then reflected in behavioral intentions, which strongly influence overt behavior.

theory of planned behavior
An extension of the theory of reasoned action, suggesting that in addition to attitudes toward a given behavior and subjective norms about it, individuals also consider their ability to perform the behavior.

implementation plan
A plan for how to implement our intentions to carry out some action.

consumed alcohol (MacDonald, Zanna, & Fong, 1995). Other behaviors, including use of the recreational drug *ecstasy*, can be predicted with careful measurement of the components suggested by these theories. For example, Orbell, Blair, Sherlock, and Conner (2001) found that having a positive attitude toward ecstasy, seeing its use as normatively accepted by one's peer group, and having perceived control over using it were all significant predictors of intentions to use this drug. In fact, attitudes, subjective norms, and intentions were all significant predictors of actual ecstasy use 2 months later.

Attitudes and Spontaneous Behavioral Reactions

Our ability to predict behavior in situations where people have the time and opportunity to reflect carefully on various possible actions that they might undertake is quite good. However, in many situations, people have to act quickly and their reactions are more spontaneous. Suppose another driver cuts in front of you on the highway without signaling. In such cases, attitudes seem to influence behavior in a more direct and seemingly automatic manner, with intentions playing a less important role. According to one theoretical view—Fazio's **attitude-to-behavior process model** (Fazio, 1990; Fazio & Roskos-Ewoldsen, 1994)—the process works as follows. Some event activates an attitude; that attitude, once activated, influences how we perceive the attitude object. At the same time, our knowledge about what's appropriate in a given situation (our knowledge of various social norms) is also activated. Together, the attitude and the previously stored information about what's appropriate or expected shape our *definition* of the event. This perception, in turn, influences our behavior. Let's consider a concrete example.

attitude-to-behavior process model
A model of how attitudes guide behavior that emphasizes the influence of attitudes and stored knowledge of what is appropriate in a given situation on an individual's definition of the present situation. This definition, in turn, influences overt behavior.

Imagine that someone cuts into your traffic lane as you are driving (see Figure 5.13). This event triggers your attitude toward people who engage in such dangerous and discourteous behavior and, at the same time, your understanding of how people are expected to behave on expressways. As a result, you perceive this behavior as non-normative, which influences your definition of and your response to that event. You might think, "Who does this person think he/she is? What nerve!" or, perhaps your response is more situational, "Gee, this person must be in a big hurry." Whichever of these interpretations of the event is given, it will shape the individual's behavior. Several studies provide support for this perspective on how attitudes can influence behavior by affecting the interpretation given to the situation.



FIGURE 5.13 Spontaneous Attitude-to-Behavior Process Effects
According to the attitude-to-behavior process view, events trigger our attitudes and, simultaneously, the appropriate norms for how people should or typically do behave in a given situation. In this case, being cut off in traffic by another driver triggers our attitudes toward such persons and our knowledge that this action is atypical. This interpretation, in turn, determines how we behave. Thus, attitudes are an important factor in shaping our overt behavior.

In short, attitudes affect our behavior through at least two mechanisms, and these operate under somewhat contrasting conditions. When we have time to engage in careful, reasoned thought, we can weigh all the alternatives and decide how we will act. Under the hectic conditions of everyday life, however, we often don't have time for this kind of deliberate weighing of alternatives, and often people's responses appear to be much faster than such deliberate thought processes can account for. In such cases, our attitudes seem to spontaneously shape our perceptions of various events—often with very little conscious cognitive processing—and

habit
Repeatedly performing a specific behavior so responses become relatively automatic whenever that situation is encountered.

thereby shapes our immediate behavioral reactions (e.g., Bargh & Chartrand, 2000; Dovidio, Brigham, Johnson, & Gaertner, 1996). To the extent that a person repeatedly performs a specific behavior—and a **habit** is formed—that person's responses may become relatively automatic whenever that same situation is encountered (Wood, Quinn, & Kashy, 2002).

KEY POINTS

- Several factors affect the strength of the relationship between attitudes and behavior; some of these relate to the situation in which the attitudes are activated, and some to aspects of the attitudes themselves.
- Attitudes seem to influence behavior through two different mechanisms. When we can give careful thought to our attitudes, *intentions* derived from our attitudes, *norms,* and *perceived control over the behavior* all predict behavior. In situations where we do not engage in such deliberate thought, attitudes may be automatically activated and influence behavior by shaping perceptions of the situation, which in turn dictate behavior.

The Fine Art of Persuasion: How Attitudes Are Changed

persuasion
Efforts to change others' attitudes through the use of various kinds of messages.

How many times in the last few days has someone tried to change your attitudes about something or other? If you stop and think for a moment, you may be surprised at the answer, for it is clear that each day we are literally bombarded with such attempts, some of which are illustrated in Figure 5.14. Billboards, television commercials, magazine ads, telemarketers, pop-up ads on your computer, and even our friends—the list of potential "would-be persuaders" seems almost endless. To what extent are such attempts at **persuasion**—efforts to change our attitudes through the use of various kinds of messages—successful? And what factors determine if they succeed or fail? Social psychologists have studied these issues for decades, and as we'll soon see, their efforts have yielded important insights into the cognitive processes that play a role in persuasion (e.g., Petty et al., 2003; Wegener & Carlston, 2005).

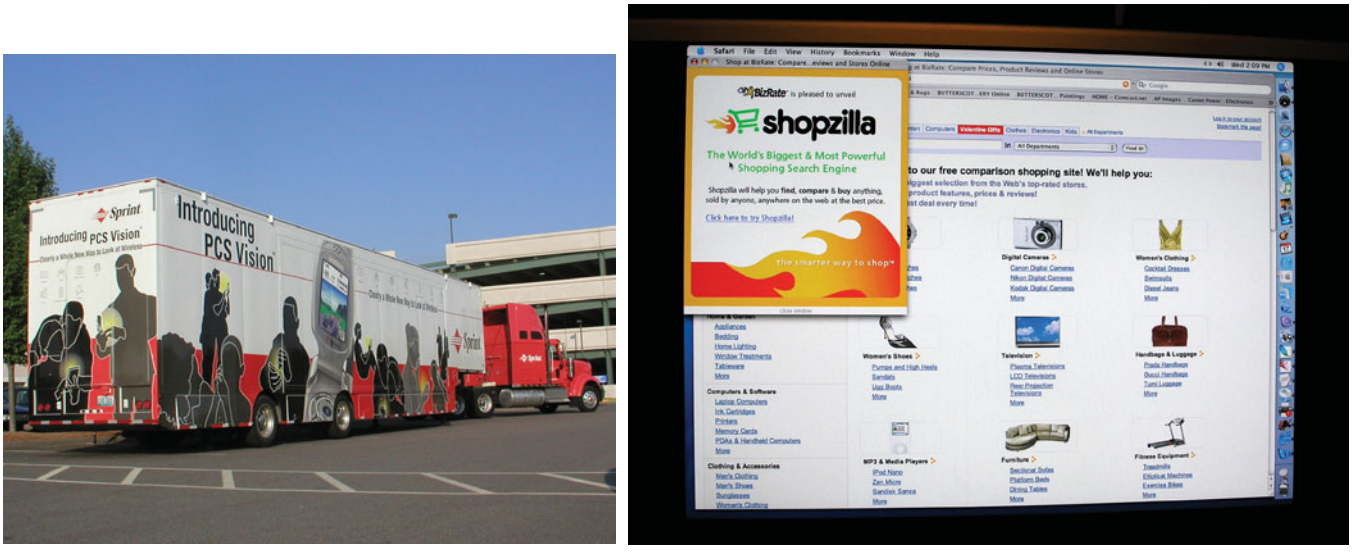


FIGURE 5.14 Persuasion: A Part of Daily Life
Each day we are bombarded with dozens of messages designed to change our attitudes or our behavior. Clearly, if they weren't effective some of the time, advertisers would not pay the sums that they do for these opportunities to try and persuade us to buy what they are promoting.

Persuasion: Communicators, Messages, and Audiences

Early research efforts aimed at understanding persuasion involved the study of the following elements: some *source* directs some type of *message* to some person or group of people (the *audience*). Persuasion research conducted by Hovland, Janis, and Kelley (1953) focused on these key elements, asking: "*Who* says *what* to *whom* with what effect?" This approach yielded a number of important findings, with the following being the most consistently obtained.

- Communicators who are *credible*—who seem to know what they are talking about or who are expert with respect to the topics or issues they are presenting—are more persuasive than those who are seen as lacking expertise. For instance, in a famous study on this topic, Hovland and Weiss (1951) asked participants to read communications dealing with various issues (e.g., atomic submarines, the future of movie theaters—remember, this was back in 1950!). The supposed source of these messages was varied so as to be high or low in credibility. For instance, for atomic submarines, a highly credible source was the famous scientist Robert J. Oppenheimer, while the low-credibility source was *Pravda*, the newspaper of the Communist party in the Soviet Union (notice how the credible source was an ingroup member, but the low-redibility source for these American participants was an outgroup source). Participants expressed their attitudes toward these issues a week before the experiment, and then immediately after receiving the communications. Those who were told that the source of the messages they read was a highly credible ingroup member showed significantly greater attitude change than those who thought the message was from the outgroup, which lacked trustworthiness and credibility. Indeed, members of our own group are typically seen as more credible and therefore are likely to influence us more than those with whom we do not share a group membership and with whom we might even *expect* to disagree (Turner, 1991).

  Communicators can, though, lose their credibility and therefore their ability to persuade. One means by which credibility can be undermined is if you learn that a communicator has a personal stake (financial or otherwise) in persuading you to adopt a particular position. Consequently, communicators are seen as most credible and therefore persuasive when they are perceived as arguing against their self-interests (Eagly, Chaiken, & Wood, 1981).

- Communicators who are physically attractive are more persuasive than communicators who are not attractive (Hovland & Weiss, 1951). Frequently, as shown in Figure 5.15, advertisers who use attractive models are attempting to suggest to us that if we buy their product, we too will be perceived as attractive. Another way that communicators can be seen as attractive is via their perceived likeability (Eagly & Chaiken, 1993). We are more likely to be persuaded by a communicator we like than one we dislike. This is one reason why famous sports figures such as Kobe Bryant, musicians such as Beyoncé



FIGURE 5.15 Role of Attractiveness in Persuasion: Can the Same Person Persuade Us to Buy Different Kinds of Products?

Research reveals that we are more persuaded by someone we view as attractive and like. In fact, actresses such as Catherine Zeta-Jones shown here are selected to be spokesperson for many different products—both those that are beauty-relevant (cosmetics, jewelry) and those that are not (cell phones).

Knowles, and actresses such as Catherine Zeta-Jones are selected as spokespeople for various products—we already like them so are more readily persuaded by them.

- Messages that do not appear to be designed to change our attitudes are often more successful than those that seem to be designed to achieve this goal (Walster & Festinger, 1962). Indeed, a meta-analysis of the existing research on this issue indicates that forewarning does typically lessen the extent to which attitude change occurs (Benoit, 1998). So, simply knowing that a sales pitch is coming your way undermines its persuasiveness.
- One approach to persuasion that has received considerable research attention is the effect of **fear appeals**—messages that are intended to arouse fear in the recipient. For example, Janis and Feshbach (1953) gave people one of three messages about the tooth decay that can result from not brushing one's teeth. They found that the mild fear-inducing message resulted in the greatest subsequent tooth brushing, while the most fear-inducing message resulted in the least increase in brushing. When the message is sufficiently fear arousing that people genuinely feel threatened, they are likely to argue against the threat, or else dismiss its applicability to themselves (Liberman & Chaiken, 1992; Taylor & Shepperd, 1998). Figure 5.16 illustrates some of the gruesome fear-based ads that have been used in an attempt to frighten people about the consequences if they fail to change their behavior. Despite the long-standing use of such fear-based messages, a recent meta-analysis of studies examining the role of fear in persuasion finds that they are not generally effective at changing people's health-related behaviors (de Hoog, Stroebe, & de Wit, 2007).

fear appeals
Attempting to change people's behaviors by use of a message that induces fear.



FIGURE 5.16 Using Fear to Encourage Change
Many messages use frightening images in an attempt to "scare people" into changing their attitudes and behavior, including the sorts of warnings illustrated here that are aimed at getting people to stop smoking and behave in environmentally friendly ways to mitigate climate change.

Might inducing more moderate levels of fear work better? There is some evidence that this is the case—but it needs to be paired with specific methods of behavioral change that will allow the negative consequences to be avoided (Petty, 1995). If people do not know how to change, or do not believe that they can succeed in doing so, fear will do little except induce avoidance and defensive responses.

Research findings (Broemer, 2004) suggest that health messages

of various sorts can be more effective if they are framed in a positive manner (e.g., how to attain good health) rather than in a negative manner (e.g., risks and the undesirable consequences that can follow from a particular behavior). For example, any health message can be framed positively as "Do this and you will feel better." Negative framing of the same message might be "If you don't do this, you will shorten your life." The point is that the same health information can be framed in terms of potential benefits of taking a particular action or in terms of the negative consequences that will ensue if you don't take that action.

Positively framed messages are often *more* effective persuasion devices than fear appeals. Consider how message framing and perceived risk of having a serious outcome befall the self can affect persuasion following exposure to a message designed to encourage low-income ethnic minority women to be tested for HIV (Apanovitch, McCarthy, & Salovey, 2003). Those women who perceived themselves as unlikely to test positive for HIV were more likely to be persuaded to be tested (and they actually got tested) when the message was framed in terms of the gains to be had by doing so (e.g., "The peace of mind you'll get or you won't have to worry that you could spread the virus") than when the message was framed in terms of potential losses they would otherwise experience (e.g., "You won't have peace of mind or you could spread the virus unknowingly to those you care about"). Positive framing can be effective in inducing change—especially when individuals fail to perceive themselves as especially at risk.

Early research on persuasion certainly provided important insights into the factors that influence persuasion. What such work did *not* do, however, was offer a comprehensive account of *how* persuasion occurs. For instance, why, precisely, are highly credible or attractive communicators more effective in changing attitudes than less credible or attractive ones? Why might positive message framing (rather than negative, fear-based) produce more attitude change? In recent years, social psychologists have recognized that to answer such questions, it is necessary to carefully examine the cognitive processes that underlie persuasion—in other words, what goes on in people's minds while they listen to a persuasive message. It is to this highly sophisticated work that we turn next.

The Cognitive Processes Underlying Persuasion

What happens when you are exposed to a persuasive message—for instance, when you watch a television commercial or see ads pop up on your screen as you surf the Internet? Your first answer might be something like, "I think about what's being said," and in a sense, that's correct. But as we saw in Chapter 2 people often do the least amount of cognitive work that they can in a given situation. Indeed, people may *want* to avoid listening to such commercial messages (and thanks to DVDs and TiVo, people can sometimes skip commercials entirely!). But when you are subjected to a message, the central issue—the one that seems to provide the key to understanding the entire process of persuasion—is really, "How do we process (absorb, interpret, evaluate) the information contained in such messages?" The answer that has emerged from hundreds of separate studies is that basically, we can process persuasive messages in two distinct ways.

systematic processing
Processing of information in a persuasive message that involves careful consideration of message content and ideas.

central route to persuasion
Attitude change resulting from systematic processing of information presented in persuasive messages.

heuristic processing
Processing of information in a peruasive message that involves the use of simple rules of thumb or mental shortcuts.

peripheral route to persuasion
Attitude change that occurs in response to peripheral persuasion cues, which is often based on information concerning the expertise or status of would-be persuaders.

elaboration-likelihood model (ELM)
A theory suggesting that persuasion can occur in either of two distinct ways, differing in the amount of cognitive effort or elaboration the message receives.

SYSTEMATIC VERSUS HEURISTIC PROCESSING The first type of processing we can employ is known as **systematic processing** or the **central route to persuasion,** and it involves careful consideration of message content and the ideas it contains. Such processing requires effort, and it absorbs much of our information-processing capacity. The second approach, known as **heuristic processing** or the **peripheral route to persuasion,** involves the use of mental shortcuts such as the belief that "experts' statements can be trusted," or the idea that "if it makes me feel good, I'm in favor of it." This kind of processing requires less effort and allows us to react to persuasive messages in an automatic manner. It occurs in response to cues in the message or situation that evoke various mental shortcuts (e.g., beautiful models evoke the "What's beautiful is good and worth listening to" heuristic).

When do we engage in each of these two distinct modes of thought? Modern theories of persuasion such as the **elaboration-likelihood model** (ELM; e.g., Petty & Cacioppo,

1986; Petty et al., 2005) and the heuristic-systematic model (e.g., Chaiken, Liberman, & Eagly, 1989; Eagly & Chaiken, 1998) provide the following answer. We engage in the most effortful and systematic processing when our motivation and capacity to process information relating to the persuasive message is high. This type of processing occurs if we have a lot of knowledge about the topic, we have a lot of time to engage in careful thought, or the issue is sufficiently important to us and we believe it is essential to form an accurate view (Maheswaran & Chaiken, 1991; Petty & Cacioppo, 1990).

In contrast, we engage in the type of processing that requires less effort (heuristic processing) when we lack the ability or capacity to process more carefully (we must make up our minds very quickly or we have little knowledge about the issue) or when our motivation to perform such cognitive work is low (the issue is unimportant to us or has little potential effect on us). Advertisers, politicians, salespeople, and others wishing to change our attitudes prefer to push us into the heuristic mode of processing because, for reasons we describe later, it is often easier to change our attitudes when we think in this mode than when we engage in more careful and systematic processing. Strong arguments in favor of the position being advocated aren't needed when people do not process those arguments very carefully! The two routes to persuasion suggested by the ELM model are shown in Figure 5.17.

What role might consuming a drug like caffeine have on persuasion? The central route to persuasion works when people attend to a message and systematically process its contents. Given that caffeine intake should increase people's ability to systematically process the contents of a message, if people have the opportunity to focus on a persuasive message without being distracted, they should be persuaded more after consuming caffeine than after not consuming it. In contrast, when people are highly distracted, it should prevent them from systematically processing the message and, if caffeine works via the central route, distraction should lessen the extent to which they are persuaded. Research findings have supported these ideas: in low-distraction conditions, those who have consumed caffeine agree more with the message (they are persuaded away from their original opinion) than those who received a caffeine-free placebo. In contrast, when people are distracted and systematic processing of the message content is impossible, there is no difference in the attitudes of those who consumed caffeine and those who did not (Martin, Hamilton,

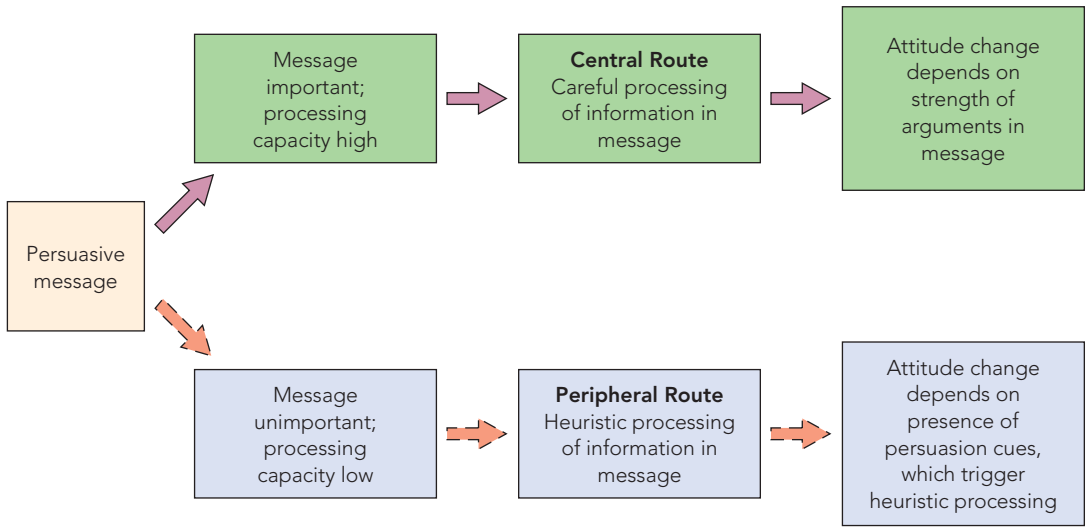


FIGURE 5.17 The ELM Model: A Cognitive Theory of Persuasion

According to the elaboration likelihood model (ELM), persuasion can occur in one of two ways. First, we can be persuaded by systematically processing the information contained in the persuasive messages (the central route), or second, by use of heuristics or mental one word shortcuts (the peripheral route). Systematic processing occurs when the message is important to us and we have the cognitive resources available to think about it carefully. Heuristic processing is most likely when the message is not important to us or we do not have the cognitive resources (or time) to engage in careful thought. *(Source: Based on suggestions by Petty & Cacioppo, 1986).*

McKimmie, Terry, & Martin, 2007). It is the increased thinking about the message when people are not distracted that can result in increased persuasion in caffeine drinkers. So, as shown in Figure 5.18, be prepared to think carefully about the messages you are exposed to when you get your next "caffeine fix"!



FIGURE 5.18 Drinking Beverages Containing Caffeine Can Increase Persuasion
Are these people, after getting a "dose" of caffeine, more likely to be persuaded by the messages they receive—than people who have not consumed caffeine? Yes, to the extent that the message is systematically processed.

The discovery of these two contrasting modes of processing—systematic versus heuristic—has provided an important key to understanding when and how persuasion occurs. For instance, when persuasive messages are not interesting or relevant to individuals, the degree of persuasion they produce is *not* strongly influenced by the strength of the arguments these messages contain. When such messages are highly relevant to individuals, however, they are much more successful in inducing persuasion when the arguments they contain *are* strong and convincing. Can you see why this so? According to modern theories such as the ELM that consider these dual pathways, when relevance is low, individuals tend to process messages through the heuristic mode, using various mental shortcuts. Thus, argument strength has little impact. In contrast, when relevance is high, they process persuasive messages more systematically and in this mode, argument strength *is* important (e.g., Petty & Cacioppo, 1990).

Similarly, the systematic versus heuristic distinction helps explain why people can be more easily persuaded when they are distracted than when they are not. Under these conditions, the capacity to process the information in a persuasive message is limited, so people adopt the heuristic mode of thought. If the message contains the "right" cues that will induce heuristic processing (e.g., communicators who are attractive or seemingly expert), persuasion may occur because people respond to these cues and *not* to the arguments being presented. In sum, the modern cognitive approach really does seem to provide the crucial key to understanding many aspects of persuasion. In the following section, "SOCIAL LIFE IN A CONNECTED WORLD: Electronic Word-of-Mouth Marketing and Persuasion," we illustrate ways that persuasion can occur over the Internet.

SOCIAL LIFE *in a* CONNECTED WORLD

Electronic Word-of-Mouth Marketing and Persuasion

Word-of-mouth marketing has been around for a long time—it simply involves providing opinions, including recommendations and general product information, in an informal, person-to-person manner (Katz & Lazarsfeld, 1955). If you have ever told someone about a good restaurant, book, movie, or made some other type of product recommendation, you've been engaged in word-of-mouth marketing. In what has come to be called eWOM (electronic word-of-mouth), Facebook, Twitter, and the many other Internet forums shown in Figure 5.19 have become means by which the transmission of word-of-mouth communications are electronically accomplished. With the increasing use of the Internet, eWOM has become a powerful and useful resource for consumers.

(continued)



FIGURE 5.19 Electronic Word-of-Mouth Marketing Forums
All of these are channels by which word-of-mouth marketing and persuasion occurs. "Friends" on Facebook, for example, comment on new products and create a "buzz" within their own social network.

I know that before I lay out $10 to see a movie, I check out what other people have to say about it on *Rotten Tomatoes* or another movie review website. But, how do we make sense of the reviews that people provide on such sites? According to Lee and Youn (2009), the more the consumer attributes a communicator's review about a product to that product's actual features, the more the consumer will perceive that communicator as credible. This leads to greater confidence in the accuracy of the review and increases the likelihood of consumer persuasion.

In the eWOM situation, there is generally less control over the flow of "advertising" in the traditional sense (Chen & Lee, 2008). Typically, in what we will call the "buzz" situation, one is tracking a conversation on Facebook or receiving tweets on Twitter, all of which involve some sort of textual material in a conversational format. We know that eWOM connects diverse individual consumers to enable conversation. This helps people utilize information from the eWOM network to make purchase decisions. But the consumer must evaluate the credibility of those who are making recommendations. In eWOM, people's questioning of the credibility of online reviews can be a real problem for marketers.

Cheung, Luo, Sia, and Chen (2009) conducted a study to investigate factors that influence credibility judgments of online consumer recommendations. Informational determinants include argument strength, source credibility, and confirmation with prior beliefs. Normative determinants include recommendation consistency and recommendation rating. Because the reader does not typically know the person who is making the recommendation, a positive response to an informational message is likely to be based on the sheer number of positive recommendations one is exposed to. Cheung et al. (2009) found that credibility is a major concern for information receivers. So recommendation ratings and recommendation consistency, the two normative components, are particularly important determinants of whether consumers are influenced.

Some information-based determinants—argument strength, source credibility, and confirmation of prior beliefs—significantly influence perceived eWOM credibility (Cheung et al., 2009). A contributor's reputation as being credible is an indicator that readers use to evaluate the eWOM message. Argument quality is also important. Readers do not simply follow comments blindly. If an online recommendation is inconsistent with the receiver's prior beliefs, the receiver will tend to suspect its credibility.

The large numbers of participants in online discussion forums allow consumers to assess consistency in eWOM messages. If a similar experience is repeatedly reported by different forum users, readers are likely to believe it. In addition, the combined rating of past readers helps users understand how other readers tend to judge an online recommendation. This increases confidence in posted reviews.

Most online retailers (e.g., Amazon.com, Zappos.com, Overstock.com, ColdwaterCreek.com) provide an opportunity for consumers to contribute after-purchase reviews. These are intended to influence consumers' online purchase intentions. Of course, consumers seek quality information and believe that at least some reviews by other consumers provide useful information. However, Zhu and Zhang (2010) found it is mostly extremely satisfied and extremely dissatisfied users who write reviews, so the consumer tends to be exposed to extreme views. Mere popularity, by itself, can be informational to buyers, and they respond to this. But these researchers found that low-selling items, or niche market products, benefited more by reviews, although they were hurt by even one bad review. Reviews are more influential when those who write them have more Internet experience. In general, retailers believe that their online reviews are very helpful to consumers.

Many consumers use the Internet to evaluate, in an informal manner, product information in what has come to be called eWOM. Facebook, Twitter, MySpace, and others are all ways in which people gain access to others' opinions. In the online environment, some consumers become inadvertent marketers and influence other consumers.



KEY POINTS

- Early research on **persuasion**—efforts to change attitudes through the use of messages—focused primarily on characteristics of the *communicator* (e.g., expertise, attractiveness), *message* (e.g., fear appeals, one-sided vs. two-sided arguments), and *audience*.
- Communicators who are deemed credible, physically attractive, and offer messages that seem not to be designed to persuade us tend to be most persuasive.
- **Fear appeals**—messages that are intended to arouse fear—if too frightening tend not to be effective. Positively framed messages are often more effective persuasion devices.
- Modern theories of persuasion include the **elaboration-likelihood model** (ELM) and the *heuristic-systematic model*. Research based on these models has sought to understand the cognitive processes that play a role in persuasion.
- Persuasive messages can be processed in two distinct ways: through **systematic processing** or **central route to persuasion,** which involves careful attention to message content, or through **heuristic processing** or **peripheral route to persuasion,** which involves the use of mental shortcuts (e.g., "experts are usually right").
- Argument strength only affects persuasion when more systematic processing is engaged, whereas peripheral cues such as features of the communicator's attractiveness or expertise only affect persuasion when more heuristic processing occurs.
- Substances such as caffeine can affect persuasion because of their effects on systematic processing of the information in a message.
- *Electronic word-of-mouth* persuasion depends on communicator credibility, consistency among reviewer recommendations, and consistency of the message with prior beliefs.

Resisting Persuasion Attempts

As we have been discussing, people can be persuaded to change their attitudes and behavior—either because they think systematically about a compelling message, or because they are influenced by more peripheral cues. Why then might people sometimes be a "tough sell" where efforts to change attitudes are concerned? The answer involves several factors that, together, enhance our ability to resist even highly skilled efforts at persuasion.

Reactance: Protecting Our Personal Freedom

Few of us like being told what to do, but in a sense that is precisely what advertisers and other would-be persuaders do. You have probably experienced another individual who increasingly pressures you to get you to change your attitude on some issue. In both of these instances, whether "public" persuaders or private ones, you are on the receiving end of threats to your freedom to decide for yourself. As a result, you may experience a growing level of annoyance and resentment. The final outcome: Not only do you resist their persuasion attempts, but you may actually lean over backward to adopt views *opposite* to those the would-be persuader wants you to adopt. Such behavior is an example of what social psychologists call **reactance**—a negative reaction to efforts by others to reduce our freedom by getting us to believe or do what *they* want (Brehm, 1966). Research indicates that in such situations, we do often change our attitudes and behavior in the opposite direction from what we are being urged to believe or to do. Indeed, when we are feeling reactance, strong arguments in favor of attitude change can increase opposition compared to moderate or weak arguments (Fuegen & Brehm, 2004). The existence of reactance is one reason why hard-sell attempts at persuasion often fail. When individuals perceive such appeals as direct threats to their

reactance
Negative reactions to threats to one's personal freedom. Reactance often increases resistance to persuasion and can even produce negative attitude change or opposite to what was intended.

personal freedom (or their image of being an independent person), they are strongly motivated to resist.

Forewarning: Prior Knowledge of Persuasive Intent

When we watch television, we fully expect there to be commercials, and we know full well that these messages are designed to persuade us to purchase various products. Similarly, we know that when we listen to a political speech that the person delivering it is attempting to persuade us to vote for him or her. Does the fact that we know in advance about the persuasive intent behind such messages help us to resist them? Research on the effects of such advance knowledge—known as **forewarning**—indicates that it does (e.g., Cialdini & Petty, 1979; Johnson, 1994). When we know that a speech or written appeal is designed to alter our views, we are often less likely to be affected by it than when we do not possess such knowledge. Why? Because forewarning influences several cognitive processes that play an important role in persuasion.

First, forewarning provides us with more opportunity to formulate *counterarguments*—those that refute the message—and that can lessen the message's impact. In addition, forewarning provides us with more time to recall relevant information that may prove useful in refuting the persuasive message. Wood and Quinn (2003) found that forewarning was generally effective at increasing resistance, and that simply *expecting* to receive a persuasive message (without actually even receiving it) can influence attitudes in a resistant direction. In many cases, then, forewarned is indeed forearmed where persuasion is concerned. But what if you are distracted between the time of the warning and receipt of the message—to such an extent that it prevents you from forming counterarguments? Research has revealed that forewarning does not prevent persuasion when people are distracted; in this case, people are no more likely to resist the message than those not forewarned of the upcoming persuasive appeal.

There are instances where forewarnings can encourage attitude shifts toward the position being advocated in a message, but this effect appears to be a temporary response to people's desire to defend their view of themselves as not gullible or easily influenced (Quinn & Wood, 2004). In this case, because people make the attitude shift before they receive the persuasive appeal, they can convince themselves that they were not in fact influenced at all! Furthermore, in such cases, distraction after forewarning has been received—which presumably would inhibit thought—has no effect on the extent to which attitudes are changed in the direction of the expected message. In this type of forewarning situation, people appear to be using a simple heuristic (e.g., I'll look stupid if I don't agree with what this expert says) and change their attitudes before they even receive the message.

Selective Avoidance of Persuasion Attempts

Still another way in which we resist attempts at persuasion is through **selective avoidance,** a tendency to direct our attention away from information that challenges our existing attitudes. Television viewing provides a clear illustration of the effects of selective avoidance. People do not simply sit in front of the television passively absorbing whatever the media decides to dish out. Instead, they channel-surf, mute the commercials, tape their favorite programs, or simply cognitively "tune out" when confronted with information contrary to their views. The opposite effect occurs as well. When we encounter information that *supports* our views, we tend to give it our full attention. Such tendencies to ignore information that contradicts our attitudes, while actively attending to information consistent with them, constitute two sides of what social psychologists term *selective exposure*. Such selectivity in what we make the focus of our attention helps ensure that many of our attitudes remain largely intact for long periods of time.

forewarning
Advance knowledge that one is about to become the target of an attempt at persuasion. Forewarning often increases resistance to the persuasion that follows.

selective avoidance
A tendency to direct attention away from information that challenges existing attitudes. Such avoidance increases resistance to persuasion.

Actively Defending Our Attitudes: Counterarguing Against the Competition

Ignoring or screening out information incongruent with our current views is certainly one way of resisting persuasion. But growing evidence suggests that in addition to this kind of passive defense of our attitudes, we also use a more active strategy as well: We actively counterargue against views that are contrary to our own (Eagly, Chen, Chaiken, & Shaw-Barnes, 1999). By doing so, it makes the opposing views more memorable than they would be otherwise, but it reduces their impact on our attitudes.

Eagly, Kulesa, Brannon, Shaw, and Hutson-Comeaux (2000) identified students as either "pro-choice" or "pro-life" in their attitudes toward abortion. These students were then exposed to persuasive messages that were either consistent with their attitudes or were contrary to their views. After hearing the messages, participants reported their attitudes toward abortion, the strength of their attitudes, and listed all the arguments in the message they could recall (a measure of memory). In addition, they listed the thoughts they had while listening to the message; this provided information on the extent to which they counterargued against the message when it was contrary to their own views.

The results indicated that the counterattitudinal message and the proattitudinal message were equally memorable. However, participants reported thinking more systematically about the counterattitudinal message, and reported having more oppositional thoughts about it—a clear sign that they were indeed counterarguing against this message. In contrast, they reported more supportive thoughts in response to the proattitudinal message. Therefore, one reason we are so good at resisting persuasion is that we not only ignore information that is inconsistent with our current views, but we also carefully process counterattitudinal input and argue actively against it. In this way, exposure to arguments opposed to our attitudes can serve to strengthen the views we already hold, making us more resistant to subsequent efforts to change them.

Individual Differences in Resistance to Persuasion

People differ in their vulnerability to persuasion (Brinol, Rucker, Tormala, & Petty, 2004). Some people may be resistant because they are motivated to engage in counterarguing; they therefore would agree with items such as "When someone challenges my beliefs, I enjoy disputing what they have to say" and "I take pleasure in arguing with those who have opinions that differ from my own." On the other hand, some people are relatively resistant to persuasion because they attempt to bolster their own beliefs when they encounter counterattitudinal messages. Those individuals would be likely to agree with items such as "When someone has a different perspective on an issue, I like to make a mental list of the reasons in support of my perspective" and "When someone gives me a point of view that conflicts with my attitudes, I like to think about why my views are right for me." To determine whether scores on these two measures of resistance to persuasion were in fact predictive of attitude change in a persuasion situation, Brinol et al. (2004) measured these self-beliefs and then gave participants an advertisement for "Brown's Department Store." These researchers found that scores on both these measures assessing different approaches to resisting persuasion predicted successful resistance to the message in the advertisement. Furthermore, the types of thoughts people have when they are confronted with a counterattitudinal message are predicted by their preference for resisting persuasion by either counterarguing or bolstering their initial attitude position. So, apparently people do know something about how they deal with attempts to persuade them, and they use their favored techniques quite effectively!

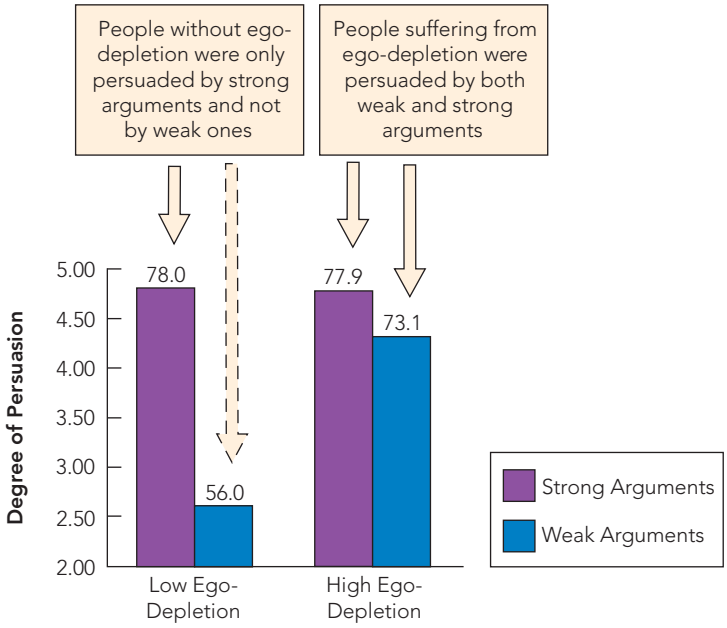


FIGURE 5.20 Evidence That Ego-Depletion Can Make Weak Ideas Persuasive

People who were not ego-depleted differentiated between weak and strong arguments, and were only persuaded by strong arguments. In contrast, people suffering from ego-depletion failed to differentiate between strong and weak arguments, and were therefore persuaded by both. (Source: Based on data from Wheeler, Brinol, & Hermann, 2007).

Ego-Depletion Can Undermine Resistance

As we just described, your ability to resist persuasion can result from successful counterarguing against a persuasive message or consciously considering why your initial attitude is better than the position you are being asked to adopt. Factors that make either of these strategies more difficult—because they undermine our ability to engage in **self-regulation**—could certainly undermine our ability to resist persuasion. To the extent that people have a limited capacity to self-regulate (i.e., to engage their will power in controlling their own thinking), prior expenditure of these limited resources could leave us vulnerable to persuasion. For example, when people are tired, have failed to self-regulate on a prior task, or otherwise are in a state of **ego-depletion,** they may simply acquiesce when confronted with a counterattitudinal message—that is, they will show attitude change.

To test this possibility, Wheeler, Brinol, and Hermann (2007) gave participants an easy or difficult first task, with the difficult task being designed to deplete their self-regulation resources. Subsequently, participants were given a weak or strong message arguing in favor of mandatory comprehensive examinations for graduation—a topic these students were initially strongly against. Did ego-depletion result in people being more persuaded by bad (weak) arguments? The answer, as shown in Figure 5.20, was a resounding yes. The weak arguments were unpersuasive to the non-ego-depleted people, but they were just as persuasive to those who were ego-depleted as were the strong arguments. For participants in the low depletion condition, strong arguments were more persuasive than weak ones, as you might expect. Examination of the participants' thoughts in response to the message verified that the low depletion participants had more favorable thoughts about the message when the arguments were strong compared to when they were weak. In contrast, the thoughts of the ego-depleted participants were equally as favorable in the strong and weak arguments case.

Recent research has confirmed too that those who have resisted a persuasive message have less ability to subsequently exert self-control (Burkley, 2008; Vohs et al., 2008; Wang, Novemsky, Dhar, & Baumeister, 2010). So not only does prior resistance deplete our self-control, which results in greater vulnerability to persuasion, but when we're depleted, we may find it more difficult to resist would-be persuaders' weak messages! Furthermore, when we are in the position of attempting to persuade others, we are more likely to be dishonest when our capacity to exert control has been depleted (Mead, Baumeister, Gino, Schweitzer, & Ariely, 2009). Participants in this research were first given a resource depleting essay to write—without using words that contained the letters A and N, or an easy one where the letters X and Z were not used. Then, participants had to find numbers in a matrix that summed to 10. Participants' performance on this task was scored by the experimenter in one condition (where cheating was not possible), or was self-scored in the other condition (where cheating was possible). Those in the resource-depleted condition and who self-scored their own performance showed the greatest cheating in reporting their performance. This research suggests that we need to beware of communicators who are the most tired when they

self-regulation
Limited capacity to engage our willpower and control our own thinking and emotions.

ego-depletion
When our capacity to self-regulate has been reduced because of prior expenditures of limited resources.

are attempting to persuade us—for they may be the most tempted to color the truth in ways that favor them over us!

KEY POINTS

- Several factors contribute to our ability to *resist persuasion*. One such factor is **reactance**—negative reactions to efforts by others to reduce or limit our personal freedom, which can produce greater overall opposition to the message content.
- Resistance to persuasion is often increased by **forewarning**—the knowledge that someone will be trying to change our attitudes—and by **selective avoidance**—the tendency to avoid exposure to information that contradicts our views.
- When we are exposed to persuasive messages that are contrary to our existing views, we actively counterargue against them. This is a critical means by which our resistance to persuasion is increased.
- There are also individual differences in the ability to resist persuasion. Those include consciously counterarguing messages we receive, and bolstering our initial attitude position when confronted with a counterattitudinal message.
- **Ego-depletion** from exerting effort on another task can undermine our ability to **self-regulate** and resist persuasion. When ego-depleted, people are equally likely to be persuaded by both strong and weak messages. As persuaders, the ego-depleted are also less likely to be honest.

Cognitive Dissonance: What Is It and How Do We Manage It?

When we first introduced the question of whether, and to what extent, attitudes and behavior are linked, we noted that in many situations, there is a sizable gap between what we feel on the inside (positive or negative reactions to some object or issue) and what we show on the outside. For instance, I have a neighbor who recently purchased a huge SUV. I have strong negative attitudes toward such giant vehicles because they get very low gas mileage, add to pollution, and block my view while driving. But when my neighbor asked how I liked her new vehicle, I hesitated and then said "Nice, very nice," with as much enthusiasm as I could muster. She is a very good neighbor who looks after my cats when I'm away, and I did not want to offend her. But I certainly felt uncomfortable when I uttered those words. Why? Because in this situation I was aware that my behavior was *not* consistent with my attitudes and this is an uncomfortable state to be in. Social psychologists term my negative reaction **cognitive dissonance** —an unpleasant state that occurs when we notice that our attitudes and our behavior are inconsistent. As you will see, when we cannot justify our attitude-inconsistent behavior (but note that I tried to do so by saying how important it was to not offend my neighbor) we may end up changing our own attitudes.

Any time you become aware of saying what you don't really believe (e.g., praise something you don't actually like "just to be polite"), make a difficult decision that requires you to reject an alternative you find attractive, or discover that something you've invested effort or money in is not as good as you expected, you are likely to experience dissonance. In all these situations, there is a gap between your attitudes and your actions, and such gaps tend to make us uncomfortable. Recent research has revealed that the discomfort associated with dissonance is reflected in elevated activity in the left front regions of our brain (Harmon-Jones, Harmon-Jones, Fearn, Sigelman, & Johnson, 2008). Most important from the present perspective, cognitive dissonance can sometimes lead us to change our own attitudes—to shift them so that they *are* consistent with our overt behavior, even in the absence of any strong external pressure to do so.

cognitive dissonance
An internal state that results when individuals notice inconsistency between two or more attitudes or between their attitudes and their behavior.

Dissonance and Attitude Change: The Effects of Induced Compliance

We can engage in attitude-discrepant behavior for many reasons, and some of these are more compelling than others. When will our attitudes change more: When there are "good" reasons for engaging in attitude-discrepant behavior or when there is little justification for doing so? As we already noted, cognitive dissonance theory argues that dissonance will be stronger when we have *few* reasons for engaging in attitude-discrepant behavior. This is so because when we have little justification and therefore cannot explain away our actions to ourselves, dissonance will be quite intense.

In the first test of this idea, participants were first asked to engage in an extremely boring series of tasks—turning pegs in a board full of holes (Festinger & Carlsmith, 1959). After the task was over, the experimenter made an unusual request: he told participants that his research assistant had not shown up that day and he asked if the participant would "fill in" by greeting the next participant, and telling that person that the task to be performed was an *interesting* one. Half of these participants were told that they would be paid $20 if they would tell this fib to the waiting participant, and the other half were told that they would receive $1 for doing so. After doing the "favor" of telling the person waiting this fib about the experiment, the participants were asked to report their own attitudes toward the boring task (i.e., rate how interesting the task was).

The participants who were paid $20 rated the task as *less* interesting than participants who were paid $1. When you were paid $20, you would have had a justification for lying, but not if you were paid $1 to tell that same lie! So, if given *insufficient justification* for your behavior, a situation that was more true in the $1 (than the $20) condition of the experiment, there is a greater need to reduce your dissonance. So, what do people do to reduce their greater dissonance in the $1 condition? They change the cognition that is causing the problem! Since, in this example, you can't change the lie you told (i.e., deny your behavior), you can decide it wasn't really a lie at all by "making" the boring task more interesting and reporting your attitude as being more positive in the $1 condition than in the $20 condition.

less-leads-to-more effect
The fact that offering individuals small rewards for engaging in counterattitudinal behavior often produces more dissonance, and so more attitude change, than offering them larger rewards.

As Figure 5.21 illustrates, cognitive dissonance theory predicts that it will be easier to change individuals' attitudes by offering them *just enough* to get them to engage in attitude-discrepant behavior. Social psychologists sometimes refer to this surprising prediction as the **less-leads-to-more effect**—less reasons or rewards for an action often leads to greater attitude change—and it has been confirmed in many studies (Harmon-Jones, 2000; Leippe & Eisenstadt, 1994). Indeed, the more money or other rewards that are offered to people for them to behave in an attitude-discrepant way provides a justification for their actions and can undermine the likelihood that attitude change will occur. Thus, coercion will serve to undermine dissonance. In addition, small rewards lead to greater attitude change primarily when people believe that they were personally responsible for both the chosen course of action and any negative effects it produced. For instance, when ordered by an authority to do a particular

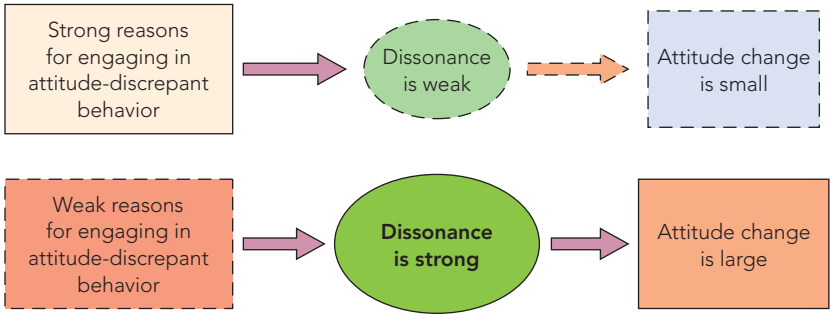


FIGURE 5.21 Why Smaller Inducements Often Lead to More Attitude Change After Attitude-Discrepant Behavior
When individuals have strong reasons for engaging in attitude-discrepant behavior, they experience relatively weak dissonance and do not change their attitudes. In contrast, when they have little apparent justification for engaging in the attitude-discrepant behavior, they will experience stronger dissonance and greater pressure to change their attitudes. The result—less justification leads to more dissonance and more change following attitude-discrepant behavior.

behavior that is inconsistent with our personal attitudes, we may not feel responsible for our actions and therefore not experience dissonance.

Alternative Strategies for Resolving Dissonance

As we have described, dissonance theory began with a very reasonable idea: People find inconsistency between their attitudes and actions uncomfortable. But is changing our attitudes the only method by which we can resolve dissonance? No, we can also alter our behavior so it is more consistent with our attitudes—for example, we could resolve to only buy organic products *in the future* and not change our "green environmental attitudes" after we've made some nonenvironmentally friendly purchase.

We can also reduce cognitive dissonance by acquiring new information (justifications) that supports our behavior. Recall our chapter opening: How might Mrs. Keech and her followers deal with their dissonance when the prophecy failed and the world did not end on the specified date? They were faced with two dissonant cognitions: "we predicted the end of the world on a certain date" and "that date has undeniably passed, and the world has not ended." After disconfirmation of the prophecy, they did *not* conclude their belief in the prophecy had been wrong, but instead the group sought to add followers in order to *reaffirm* the rightness of their beliefs. Adding followers to the group adds a consonant cognition: *great numbers* of faithful believers couldn't be wrong! Indeed, when the "end of the world" date had passed, the group reported that Earth had been spared *because* of their strong faith. By adding this belief that their faith saved Earth, these believers were able to resolve their dissonance, without changing their attitudes or behavior.

Another option for managing dissonance when inconsistency is salient involves deciding that the inconsistency actually doesn't matter! In other words, we can engage in *trivialization*—concluding that either the attitudes or behaviors in question are not important so any inconsistency between them is of no importance (Simon, Greenberg, & Brehm, 1995).

All of these strategies can be viewed as *direct* methods of dissonance reduction: They focus on the attitude–behavior discrepancy that is causing the dissonance. Research by Steele and his colleagues (e.g., Steele, 1988; Steele & Lui, 1983) indicates that dissonance can be reduced via *indirect* means. That is, although the basic discrepancy between the attitude and behavior are left intact, the unpleasant or negative feelings generated by dissonance can still be reduced by, for example, consuming alcohol. Adoption of indirect tactics to reduce dissonance is most likely when the attitude–behavior discrepancy involves *important* attitudes or self-beliefs (so trivialization isn't feasible). Under these conditions, individuals experiencing dissonance may not focus so much on reducing the gap between their attitudes and behavior, but instead on other methods that will allow them to feel good about themselves despite the gap (Steele, Spencer, & Lynch, 1993).

Specifically, people will engage in *self-affirmation*—restoring positive self-evaluations that are threatened by the dissonance (Elliot & Devine, 1994; Tesser, Martin, & Cornell, 1996). This can be accomplished by focusing on positive self-attributes—good things about oneself. For instance, when I experienced dissonance as a result of saying nice things about my neighbor's giant new SUV, even though I am strongly against such vehicles, I could remind myself that I am a considerate person. By contemplating positive aspects of the self, it can help to reduce the discomfort produced by my failure to act in a way that was consistent with my pro-environmental (and anti-SUV) attitudes. However we choose to reduce dissonance—through indirect tactics or direct strategies that are aimed at reducing the attitude–behavior discrepancy—we all find strategies to help us deal with the discomfort that comes from being aware of discrepancies between our attitudes and behavior.

When Dissonance Is a Tool for Beneficial Changes in Behavior

- People who don't wear seat belts are much more likely to die in accidents than those who do . . .
- People who smoke are much more likely to suffer from lung cancer and heart disease than those who don't . . .
- People who engage in unprotected sex are much more likely than those who engage in safe sex to contract dangerous diseases, including AIDS, as well as have unplanned pregnancies . . .

Most of us know these statements are true, and our attitudes are generally favorable toward using seat belts, quitting smoking, and engaging in safe sex (Carey, Morrison-Beedy, & Johnson, 1997). Despite having positive attitudes, they are often not translated into overt actions: Some people continue to drive without seatbelts, to smoke, and to have unprotected sex. To address these major social problems, perhaps what's needed is not so much a change in attitudes as shifts in overt behavior. Can dissonance be used to promote beneficial behavioral changes? A growing body of evidence suggests that it can (Batson, Kobrynowicz, Dinnerstein, Kampf, & Wilson, 1997; Gibbons, Eggleston, & Benthin, 1997), especially when it is used to generate feelings of **hypocrisy**—publicly advocating some attitude, and then making salient to the person that they have acted in a way that is inconsistent with their own attitudes. Such feelings might be sufficiently intense that only actions that reduce dissonance directly, by inducing behavioral change, may be effective. These predictions concerning the possibility of dissonance-induced behavior change have been tested in several studies.

hypocrisy
Publicly advocating some attitudes or behavior and then acting in a way that is inconsistent with these attitudes or behavior.



FIGURE 5.22 Indirect Route to Dissonance Reduction
When individuals are made to confront their own hypocrisy, most choose to reduce their dissonance through direct means (by changing their behavior). However, when individuals are asked to think about reasons why people in general do not act according to their beliefs, many choose to reduce dissonance via an indirect route such as donating to charity. Doing so allows people to feel better about themselves, even though their own behavior does not change.

Stone, Wiegand, Cooper, and Aronson (1997) asked participants to prepare a videotape advocating the use of condoms (safe sex) to avoid contracting AIDS. Next, participants were asked to think about reasons why they themselves hadn't used condoms in the past (*personal reasons*) or reasons why people in general sometimes fail to use condoms (*normative reasons* that didn't center on their own behavior). The researchers predicted that dissonance would be maximized in the personal reasons condition, where participants had to come face-to-face with their own hypocrisy. Then, all people in the study were given a choice between a direct means of reducing dissonance—purchasing condoms at a reduced price—or an indirect means of reducing dissonance—making a donation to a program designed to aid homeless people (see Figure 5.22). The results indicated that when participants had been asked to focus on the reasons why they didn't engage in safe sex in the past, an overwhelming majority chose to purchase condoms, suggesting that their behavior in the future will be different—the direct route to dissonance reduction. In contrast, when asked to think about reasons why people in general didn't engage in safe sex, more actually chose the indirect route to dissonance reduction—a donation to the aid-the-homeless project—and didn't change their behavior.

These findings suggest that using dissonance to make our own hypocrisy salient can indeed be a powerful tool for changing our behavior in desirable ways. For maximum effectiveness, however, such procedures must involve several elements: People must publicly advocate the desired behaviors (e.g., using condoms), they need to be induced to think about their own behavioral failures in the past, and they must be given access to direct means for reducing their dissonance (i.e., a method for changing their behavior). When these conditions are met, dissonance can bring about beneficial changes in behavior.



KEY POINTS

- **Cognitive dissonance** is an aversive state that occurs when we notice discrepancies between our attitudes and our behavior. Experiencing dissonance does indeed produce increased left frontal cortical activity and attitude change.
- Dissonance often occurs in situations involving *forced compliance,* in which we are minimally induced by external factors to say or do things that are inconsistent with our attitudes.
- Dissonance can lead to attitude change when we have reasons that are barely sufficient to get us to engage in attitude–discrepant behavior. Stronger reasons (or larger rewards) produce *less* attitude change; this is sometimes referred to as the **less-leads-to-more effect.**
- Dissonance can be reduced directly (e.g., changing our attitudes) or by adding cognitions that justify our behavior.
- Other methods for dealing with dissonance include *trivialization* and *indirect* methods such as *self-affirmation* on some other dimension.
- Dissonance induced through **hypocrisy**—inducing individuals to advocate certain attitudes or behaviors and then reminding them that their own behavior has not always been consistent with these attitudes—can be a powerful tool for inducing beneficial changes in behavior.



SUMMARY *and* REVIEW

- **Attitudes** are evaluations that can color our experience of virtually any aspect of the world. Often, attitudes are **explicit**—consciously accessible and easy to report. But attitudes can also be **implicit,** and therefore not consciously accessible or controllable. Attitudes are often acquired from other people through **social learning.** Such learning can involve *classical* **conditioning, instrumental conditioning, or observational learning.** In fact, attitudes can be formed via **subliminal conditioning**—which occurs in the absence of conscious awareness of the stimuli involved—and **mere exposure.** Attitudes are also formed on the basis of **social comparison**—our tendency to compare ourselves with others to determine whether our view of social reality is or is not correct. In order to be similar to others we like, we accept the attitudes that they hold, to the extent that we identify with that group. As we move into new **social networks,** attitudes can shift rapidly as a means of fitting in when those networks consist of people holding diverging attitudes.

- Several factors affect the strength of the relationship between attitudes and behavior. *Situational constraints* may prevent us from expressing our attitudes overtly—including concerns about what others may think of us. People often show **pluralistic ignorance**—erroneously believing that others have different attitudes than we do, which can limit our willingness to express our attitudes in public. Several aspects of attitudes themselves also moderate the attitude–behavior link. These include factors related to attitude strength, including the *extremity* of our attitude position, the *certainty* with which our attitudes are held, and whether we have *personal experience* with the attitude object. All of these factors can make our attitudes more accessible, and therefore likely to guide our behavior.

- Attitudes can influence behavior through two different mechanisms. According to the **theory of reasoned action** and **theory of planned behavior,** when we can give careful thought to our attitudes, intentions derived from our attitudes strongly predict behavior. According to the **attitude-to-behavior process model,** in situations where our behavior is more spontaneous and we don't engage in such deliberate thought, attitudes influence behavior by shaping our perception and interpretation of the situation.

- Early research on **persuasion**—efforts to change attitudes through the use of messages—focused primarily on the *source,* the *message,* and the *audience.* **Fear appeals** are limited in their ability to produce health behavior change. More recent research has sought to understand the cognitive processes that play a role in persuasion. Such research suggests that we process persuasive messages in two distinct ways: through **systematic processing,** which involves careful attention to message content, or through **heuristic processing,** which involves the use of mental shortcuts (e.g., "experts are usually right"). Consuming caffeine increases the extent to which people are persuaded by increasing their ability to systematically process the message contents. Normative and informational cues affect the extent

to which we are persuaded by *electronic word-of-mouth* communications.

- Several factors contribute to such people's ability to resist persuasion. One factor is **reactance**—negative reactions to efforts by others to reduce or limit our personal freedom. When people feel reactance, they often change their attitudes in the *opposite* direction from that advocated. This is one reason why the "hard-sell" can be counterproductive. Resistance to persuasion is often increased by **forewarning**—the knowledge that someone is trying to change our attitudes. This typically gives us a chance to counterargue against the expected persuasive appeal, and thereby resist the message content when it is presented. Forewarning does not prevent persuasion though when people are distracted and therefore unable to expend effort refuting the message in advance.

- People also maintain their current attitudes by **selective avoidance**—the tendency to overlook or disregard information that contradicts our existing views. Likewise, people give close attention to information that supports their views, and by means of *selective exposure* will actively seek out information that is consistent with their existing attitudes.

- When exposed to information that is inconsistent with our views, we can actively counterargue against them. The more people have oppositional thoughts when exposed to a counterattitudinal message, the more they are able to resist being persuaded by it. In a sense, people provide their own defense against persuasion attempts. People also differ in their vulnerability to persuasion. Some people are aware that they use counterarguing and others know they attempt to bolster their original views when they are in persuasion situations.

- Our ability to resist persuasion can depend on our own psychological state—whether we are **ego-depleted** or not. When ego-depleted, people experience greater difficulty **self-regulating,** which can undermine our ability to resist persuasion. Research has revealed that when people are ego-depleted, they do not differentiate between messages with strong and weak arguments and are equally persuaded by both. In contrast, when ego-depletion is low, people are not persuaded by weak arguments, only by strong arguments.

- **Cognitive dissonance** is an unpleasant state that occurs when we notice discrepancies between our attitudes and our behavior. Dissonance is aversive and attempts to resolve it are reflected in increased cortical activity. Festinger and Carlsmith's (1959) classic study illustrated that dissonance is stronger when we have little justification for our attitude-inconsistent behavior. In contrast, stronger reasons (or larger rewards) can produce *less* attitude change—the **less-leads-to-more effect**—because the person feels justified in their attitude–inconsistent behavior in that case.

- Dissonance often occurs in situations involving *forced compliance*—ones in which we are induced by external factors to say or do things that are inconsistent with our true attitudes. In such situations, attitude change is maximal when we have reasons that are barely sufficient to get us to engage in attitude–discrepant behavior. Other means of coping with dissonance, besides changing our attitudes, include adding justifications, trivialization, or concluding that the inconsistency doesn't matter. Dissonance can also be dealt with by use of indirect strategies; that is, to the extent that the self can be affirmed by focusing on some other positive feature of the self, then dissonance can be reduced without changing one's attitudes. Dissonance that is induced by making us aware of our own **hypocrisy** can result in behavioral changes.

KEY TERMS

attitude (p. 140)
attitude-to-behavior process model (p. 157)
central route to persuasion (p. 161)
classical conditioning (p. 144)
cognitive dissonance (p. 169)
conditioned stimulus (p. 144)
ego-depletion (p. 168)
elaboration-likelihood model (ELM) (p. 161)
explicit attitudes (p. 141)
fear appeals (p. 160)
forewarning (p. 166)
habit (p. 158)
heuristic processing (p. 161)
hypocrisy (p. 172)
illusion of truth effect (p. 146)
implicit attitudes (p. 141)
implementation plan (p. 156)
instrumental conditioning (p. 146)
less-leads-to-more effect (p. 170)
mere exposure (p. 145)
observational learning (p. 148)
peripheral route to persuasion (p. 161)
persuasion (p. 158)
pluralistic ignorance (p. 150)
reactance (p. 165)
reference groups (p. 148)
selective avoidance (p. 166)
self-regulation (p. 168)
social comparison (p. 148)
social learning (p. 144)
social networks (p. 146)
subliminal conditioning (p. 145)
systematic processing (p. 161)
theory of planned behavior (p. 156)
theory of reasoned action (p. 156)
unconditioned stimulus (p. 144)